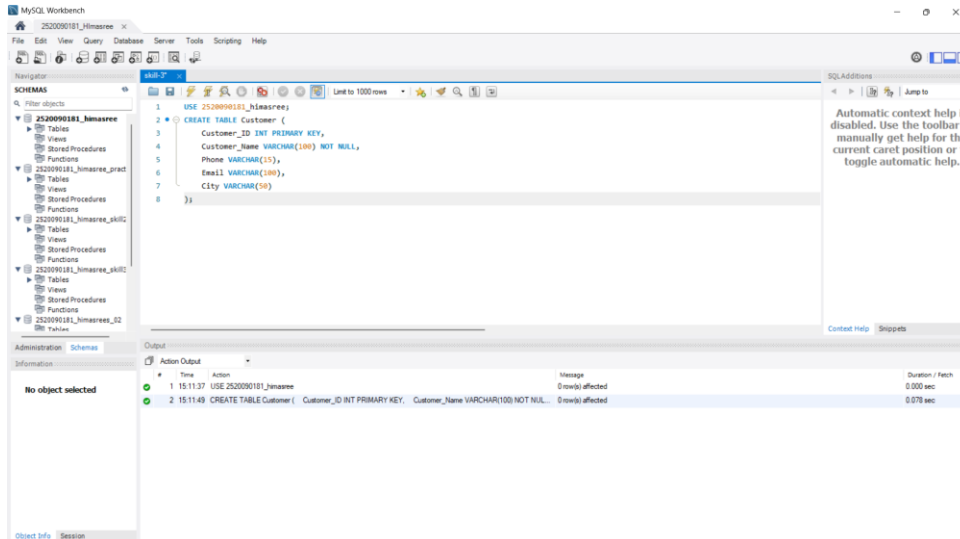
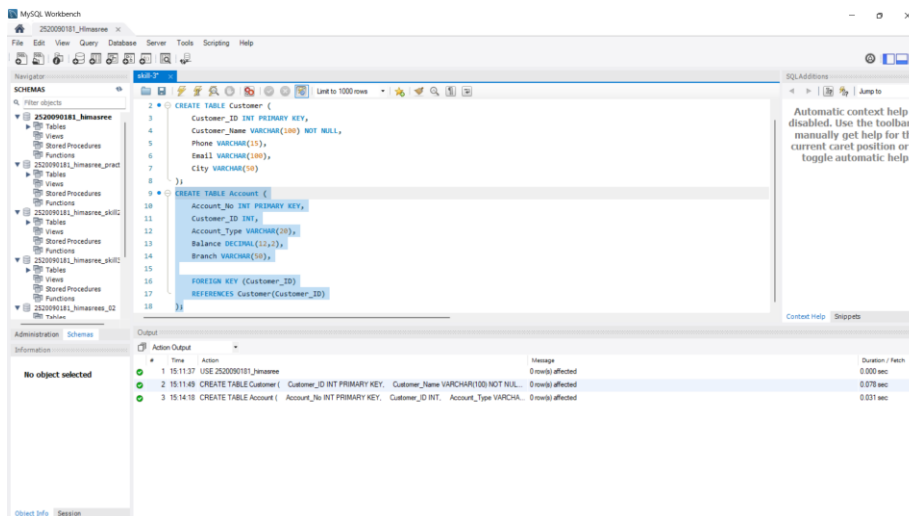


VIEWS

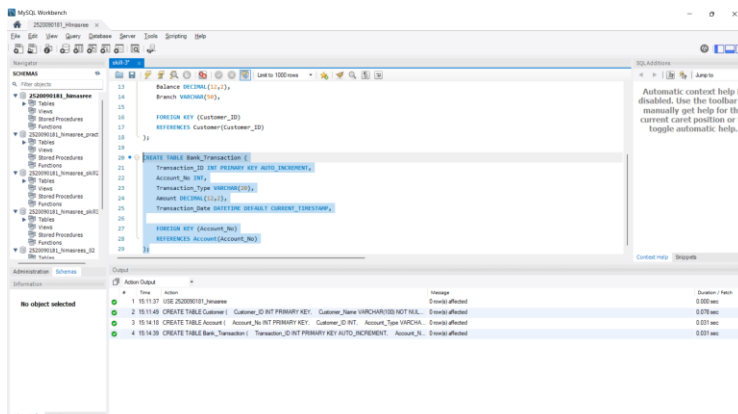
1. CREATE DATABASE / 2. CREATE CUSTOMER TABLE



3. CREATE ACCOUNT TABLE



4. CREATE BANK TRANSACTION TABLE



5. CREATE LOAN TABLE

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following SQL code:

```
CREATE TABLE Loan (  
  Loan_ID INT PRIMARY KEY,  
  Customer_ID INT,  
  Loan_Type VARCHAR(10),  
  Loan_Amount DECIMAL(12,2),  
  Interest_Rate DECIMAL(5,2),  
  FOREIGN KEY (Customer_ID)  
  REFERENCES Customer(Customer_ID)  
);
```

The Output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	15:11:37	USE 2520090181_himansree	0 row(s) affected	0.000 sec
2	15:11:49	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL, ...)	0 row(s) affected	0.078 sec
3	15:14:18	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHAR(10), ...)	0 row(s) affected	0.031 sec
4	15:14:39	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_No INT, ...)	0 row(s) affected	0.031 sec
5	15:15:12	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(10), ...)	0 row(s) affected	0.032 sec

6-9. INSERT CUSTOMER DATA / ACCOUNT DATA / TRANSACTION DATA / LOAN DATA

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following SQL code:

```
INSERT INTO Customer (  
  Customer_ID, Customer_Name, Phone, Email, City  
)  
VALUES  
(181, 'Ravi Kumar', '9876543210', 'ravi@gmail.com', 'Hyderabad'),  
(182, 'Priya Sharma', '9876543211', 'priya@gmail.com', 'Vijayawada'),  
(183, 'Arjun Reddy', '9876543212', 'arjun@gmail.com', 'Bangalore'),  
(184, 'Sneha Rao', '9876543213', 'sneha@gmail.com', 'Chennai'),  
(185, 'Kiran Kumar', '9876543214', 'kiran@gmail.com', 'Hyderabad');
```

The Output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	15:11:37	USE 2520090181_himansree	0 row(s) affected	0.000 sec
2	15:11:49	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL, ...)	0 row(s) affected	0.078 sec
3	15:14:18	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHAR(10), ...)	0 row(s) affected	0.031 sec
4	15:14:39	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_No INT, ...)	0 row(s) affected	0.031 sec
5	15:15:12	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(10), ...)	0 row(s) affected	0.032 sec
6	15:15:42	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (181, 'Ravi Kumar', '9876543210', ...)	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following SQL code:

```
INSERT INTO Loan (  
  Loan_ID, Customer_ID, Loan_Type,  
  Loan_Amount, Interest_Rate  
)  
VALUES  
(181, 181, 'Home Loan', 500000, 7.5),  
(182, 182, 'Education Loan', 300000, 8.5),  
(183, 183, 'Car Loan', 80000, 9.5),  
(184, 184, 'Personal Loan', 10000, 10.5),  
(185, 185, 'Home Loan', 400000, 7.2),  
(186, 186, 'Car Loan', 90000, 9.5);
```

The Output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
7	15:16:22	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (1001, 181, ...)	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec
8	15:16:22	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (1002, 182, ...)	Error Code: 1064 You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '1002' at line 1	0.000 sec
9	15:21:25	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (1001, 181, ...)	Error Code: 1062 Duplicate entry '1001' for key 'Account PRIMARY'	0.016 sec
10	15:21:34	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (181, 'Ravi Kumar', '9876543210', ...)	Error Code: 1062 Duplicate entry '181' for key 'Customer PRIMARY'	0.000 sec
11	15:21:58	USE 2520090181_himansree	0 row(s) affected	0.000 sec
12	15:22:01	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL, ...)	Error Code: 1058 Table 'Customer' already exists	0.015 sec
13	15:22:09	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHAR(10), ...)	Error Code: 1058 Table 'Account' already exists	0.000 sec
14	15:22:13	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_No INT, ...)	Error Code: 1058 Table 'Bank_Transaction' already exists	0.000 sec
15	15:22:25	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (1001, 181, ...)	Error Code: 1062 Duplicate entry '1001' for key 'Account PRIMARY'	0.000 sec
16	15:22:44	INSERT INTO Bank_Transaction (Account_No, Transaction_ID, Transaction_Type, Amount) VALUES (1001, DEPOSIT, 100, ...)	11 row(s) affected Records: 11 Duplicates: 0 Warnings: 0	0.016 sec
17	15:23:04	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES (181, 181, ...)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.000 sec

10. DISPLAY ORIGINAL TABLES

MySQL Workbench

2520090181_himansree

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

Schemas

2520090181_himansree

Tables

Views

Stored Procedures

Functions

2520090181_himansree_grant

Tables

Views

Stored Procedures

Functions

2520090181_himansree_skill

Tables

Views

Stored Procedures

Functions

2520090181_himansree_skill2

Tables

Views

Stored Procedures

Functions

2520090181_himansree_S2

Tables

Views

Stored Procedures

Functions

SQL Editor

Limit to 1000 rows

SELECT * FROM Customer;

Result Grid

Filter Rows

Export

Wrap Cell Content

Loan_ID Customer_ID Loan_Type Loan_Amount Interest_Rate

101 101 Home Loan 500000.00 7.50

102 102 Education Loan 200000.00 4.50

103 103 Car Loan 80000.00 8.20

104 104 Personal Loan 30000.00 10.50

105 105 Home Loan 400000.00 2.50

106 106 Car Loan 90000.00 8.50

107 107 Business Loan 300000.00 9.00

108 108 Personal Loan 60000.00 10.50

Output

Action Output

Time Action Message Duration / Fetch

11 15:21:50 USE 2520090181_himansree 0 rows affected 0.000 sec

12 15:22:01 CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL, Error Code: 1050. Table 'Customer' already exists. 0.016 sec

13 15:22:09 CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHAR(10) NOT NULL, Error Code: 1050. Table 'Account' already exists. 0.000 sec

14 15:22:13 CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_No INT, Error Code: 1050. Table 'Bank_Transaction' already exists. 0.000 sec

15 15:22:25 INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, Error Code: 1062. Duplicate entry '10001' for key 'Account PRIMARY'. 0.000 sec

16 15:22:44 INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10, Error Code: 1062. Duplicate entry '10001' for key 'Bank_Transaction PRIMARY'. 0.016 sec

17 15:23:04 INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES (501, 101, Error Code: 1062. Duplicate entry '501' for key 'Loan PRIMARY'. 0.000 sec

18 15:25:03 SELECT * FROM Customer LIMIT 0, 1000 10 rows returned 0.000 sec / 0.000 sec

19 15:25:03 SELECT * FROM Account LIMIT 0, 1000 10 rows returned 0.000 sec / 0.000 sec

20 15:25:03 SELECT * FROM Bank_Transaction LIMIT 0, 1000 11 rows returned 0.000 sec / 0.000 sec

21 15:25:03 SELECT * FROM Loan LIMIT 0, 1000 8 rows returned 0.000 sec / 0.000 sec

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

11. CREATE A VIEW FOR ALL CUSTOMERS

MySQL Workbench

2520090181_himansree

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

Schemas

2520090181_himansree

Tables

Views

Stored Procedures

Functions

2520090181_himansree_grant

Tables

Views

Stored Procedures

Functions

2520090181_himansree_skill

Tables

Views

Stored Procedures

Functions

2520090181_himansree_skill2

Tables

Views

Stored Procedures

Functions

2520090181_himansree_S2

Tables

Views

Stored Procedures

Functions

SQL Editor

Limit to 1000 rows

CREATE VIEW Customer_View AS
SELECT *
FROM Customer;

Result Grid

Filter Rows

Export

Wrap Cell Content

Customer_ID Customer_Name Email City

101 Ravi Kumar 9876543210 ravi@gmail.com Hyderabad

102 Priya Sharma 9876543211 priya@gmail.com Vijayawada

103 Arjun Reddy 9876543212 arjun@gmail.com Bangalore

104 Sneha Rao 9876543213 sneha@gmail.com Chennai

105 Kran Kumar 9876543214 kran@gmail.com Hyderabad

106 Anil Kumar 9876543215 anil@gmail.com Delhi

107 Meena Reddy 9876543216 meena@gmail.com Mumbai

108 Rahul Sharma 9876543217 rahul@gmail.com Pune

Output

Action Output

Time Action Message Duration / Fetch

18 15:25:03 SELECT * FROM Customer LIMIT 0, 1000 10 rows returned 0.000 sec / 0.000 sec

19 15:25:03 SELECT * FROM Account LIMIT 0, 1000 10 rows returned 0.000 sec / 0.000 sec

20 15:25:03 SELECT * FROM Bank_Transaction LIMIT 0, 1000 11 rows returned 0.000 sec / 0.000 sec

21 15:25:03 SELECT * FROM Loan LIMIT 0, 1000 8 rows returned 0.000 sec / 0.000 sec

22 15:26:34 CREATE VIEW Customer_View AS SELECT * FROM Customer 0 rows affected 0.016 sec

23 15:26:34 DISPLAY VIEW SELECT * FROM Customer_View Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'SELECT * FROM Customer_View' at line 1. 0.000 sec

24 15:30:40 CREATE VIEW Customer_View AS SELECT * FROM Customer Error Code: 1050. Table 'Customer_View' already exists. 0.000 sec

25 15:34:25 CREATE VIEW Customer_View AS SELECT * FROM Customer Error Code: 1050. Table 'Customer_View' already exists. 0.000 sec

26 15:34:39 DROP VIEW IF EXISTS Customer_View 0 rows affected 0.016 sec

27 15:34:42 CREATE VIEW Customer_View AS SELECT * FROM Customer 0 rows affected 0.016 sec

28 15:34:52 SELECT * FROM Customer_View LIMIT 0, 1000 10 rows returned 0.000 sec / 0.000 sec

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

12. CREATE A VIEW FOR CUSTOMER BASIC INFORMATION

MySQL Workbench

2520090181_himansree

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

Schemas

2520090181_himansree

Tables

Views

Stored Procedures

Functions

2520090181_himansree_grant

Tables

Views

Stored Procedures

Functions

2520090181_himansree_skill

Tables

Views

Stored Procedures

Functions

2520090181_himansree_skill2

Tables

Views

Stored Procedures

Functions

2520090181_himansree_S2

Tables

Views

Stored Procedures

Functions

SQL Editor

Limit to 1000 rows

CREATE VIEW Customer_Basic_View AS
SELECT
Customer_ID,
Customer_Name,
City
FROM Customer;

Result Grid

Filter Rows

Export

Wrap Cell Content

Customer_ID Customer_Name City

101 Ravi Kumar Hyderabad

102 Priya Sharma Vijayawada

103 Arjun Reddy Bangalore

104 Sneha Rao Chennai

105 Kran Kumar Hyderabad

106 Anil Kumar Delhi

107 Meena Reddy Mumbai

108 Rahul Sharma Pune

Output

Action Output

Time Action Message Duration / Fetch

20 15:25:03 SELECT * FROM Bank_Transaction LIMIT 0, 1000 11 rows returned 0.000 sec / 0.000 sec

21 15:25:03 SELECT * FROM Loan LIMIT 0, 1000 8 rows returned 0.000 sec / 0.000 sec

22 15:26:34 CREATE VIEW Customer_View AS SELECT * FROM Customer 0 rows affected 0.016 sec

23 15:26:34 DISPLAY VIEW SELECT * FROM Customer_View Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'SELECT * FROM Customer_View' at line 1. 0.000 sec

24 15:30:40 CREATE VIEW Customer_View AS SELECT * FROM Customer Error Code: 1050. Table 'Customer_View' already exists. 0.000 sec

25 15:34:25 CREATE VIEW Customer_View AS SELECT * FROM Customer Error Code: 1050. Table 'Customer_View' already exists. 0.000 sec

26 15:34:39 DROP VIEW IF EXISTS Customer_View 0 rows affected 0.016 sec

27 15:34:42 CREATE VIEW Customer_View AS SELECT * FROM Customer 0 rows affected 0.016 sec

28 15:34:52 SELECT * FROM Customer_View LIMIT 0, 1000 10 rows returned 0.000 sec / 0.000 sec

29 15:35:35 CREATE VIEW Customer_Basic_View AS SELECT Customer_ID, Customer_Name, City FROM Customer 0 rows affected 0.047 sec

30 15:35:45 SELECT * FROM Customer_Basic_View LIMIT 0, 1000 10 rows returned 0.000 sec / 0.000 sec

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

13. CREATE A VIEW FOR ACCOUNT INFORMATION

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following SQL code:

```
153 SELECT
154     Account_No,
155     Account_Type,
156     Balance,
157     Branch
158 FROM Account;
159 SELECT * FROM Account_View;
```

The Result Grid displays the following data:

Account_No	Account_Type	Balance	Branch
10001	Savings	90000.00	Hyderabad
10002	Savings	75000.00	Vijayawada
10003	Current	120000.00	Bangalore
10004	Savings	40000.00	Chennai
10005	Current	90000.00	Hyderabad
10006	Savings	60000.00	Delhi
10007	Current	150000.00	Mumbai
10008	Savings	38000.00	Pune

The Output pane shows the execution log, including the creation of the view:

```
22 15:26:34 CREATE VIEW Customer_View AS SELECT * FROM Customer 0 rows(s) affected 0.016 sec
23 15:26:34 DISPLAY VIEW SELECT * FROM Customer_View Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'SELECT * FROM Customer_View' at line 1 0.000 sec
24 15:30:48 CREATE VIEW Customer_View AS SELECT * FROM Customer Error Code: 1050. Table 'Customer_View' already exists 0.000 sec
25 15:34:25 CREATE VIEW Customer_View AS SELECT * FROM Customer Error Code: 1050. Table 'Customer_View' already exists 0.000 sec
26 15:34:39 DROP VIEW IF EXISTS Customer_View 0 rows(s) affected 0.016 sec
27 15:34:42 CREATE VIEW Customer_View AS SELECT * FROM Customer 0 rows(s) affected 0.016 sec
28 15:34:52 SELECT * FROM Customer_View LIMIT 0, 1000 10 rows(s) returned 0.000 sec / 0.000 sec
29 15:35:35 CREATE VIEW Customer_Basic_View AS SELECT Customer_ID, Customer_Name, City FROM Customer 0 rows(s) affected 0.047 sec
30 15:35:45 SELECT * FROM Customer_Basic_View LIMIT 0, 1000 10 rows(s) returned 0.000 sec / 0.000 sec
31 15:36:18 CREATE VIEW Account_View AS SELECT Account_No, Account_Type, Balance, Branch FROM ... 0 rows(s) affected 0.031 sec
32 15:36:28 SELECT * FROM Account_View LIMIT 0, 1000 10 rows(s) returned 0.000 sec / 0.000 sec
```

14. CREATE A VIEW FOR SAVINGS ACCOUNTS

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following SQL code:

```
161 CREATE VIEW Savings_Account_View AS
162 SELECT *
163 FROM Account
164 WHERE Account_Type = 'Savings';
165
166 SELECT * FROM Savings_Account_View;
```

The Result Grid displays the following data:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	90000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10004	104	Savings	40000.00	Chennai
10006	106	Savings	60000.00	Delhi
10008	108	Savings	35000.00	Pune
10009	109	Savings	85000.00	Hyderabad

The Output pane shows the execution log, including the creation of the view:

```
24 15:32:48 CREATE VIEW Customer_View AS SELECT * FROM Customer 0 rows(s) affected 0.000 sec
25 15:34:25 CREATE VIEW Customer_View AS SELECT * FROM Customer Error Code: 1050. Table 'Customer_View' already exists 0.000 sec
26 15:34:39 DROP VIEW IF EXISTS Customer_View 0 rows(s) affected 0.016 sec
27 15:34:42 CREATE VIEW Customer_View AS SELECT * FROM Customer 0 rows(s) affected 0.016 sec
28 15:34:52 SELECT * FROM Customer_View LIMIT 0, 1000 10 rows(s) returned 0.000 sec / 0.000 sec
29 15:35:35 CREATE VIEW Customer_Basic_View AS SELECT Customer_ID, Customer_Name, City FROM Customer 0 rows(s) affected 0.047 sec
30 15:35:45 SELECT * FROM Customer_Basic_View LIMIT 0, 1000 10 rows(s) returned 0.000 sec / 0.000 sec
31 15:36:18 CREATE VIEW Account_View AS SELECT Account_No, Account_Type, Balance, Branch FROM ... 0 rows(s) affected 0.031 sec
32 15:36:28 SELECT * FROM Account_View LIMIT 0, 1000 10 rows(s) returned 0.000 sec / 0.000 sec
33 15:36:56 CREATE VIEW Savings_Account_View AS SELECT * FROM Account WHERE Account_Type = 'Savings' 0 rows(s) affected 0.015 sec
34 15:37:08 SELECT * FROM Savings_Account_View LIMIT 0, 1000 6 rows(s) returned 0.000 sec / 0.000 sec
```

15. CREATE A VIEW FOR CURRENT ACCOUNTS

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following SQL code:

```
167
168 CREATE VIEW Current_Account_View AS
169 SELECT *
170 FROM Account
171 WHERE Account_Type = 'Current';
172
173 SELECT * FROM Current_Account_View;
```

The Result Grid displays the following data:

Account_No	Customer_ID	Account_Type	Balance	Branch
10003	103	Current	120000.00	Bangalore
10005	105	Current	90000.00	Hyderabad
10007	107	Current	150000.00	Mumbai
10010	110	Current	110000.00	Vijayawada

The Output pane shows the execution log, including the creation of the view:

```
26 15:34:39 DROP VIEW IF EXISTS Customer_View 0 rows(s) affected 0.016 sec
27 15:34:42 CREATE VIEW Customer_View AS SELECT * FROM Customer 0 rows(s) affected 0.016 sec
28 15:34:52 SELECT * FROM Customer_View LIMIT 0, 1000 10 rows(s) returned 0.000 sec / 0.000 sec
29 15:35:35 CREATE VIEW Customer_Basic_View AS SELECT Customer_ID, Customer_Name, City FROM Customer 0 rows(s) affected 0.047 sec
30 15:35:45 SELECT * FROM Customer_Basic_View LIMIT 0, 1000 10 rows(s) returned 0.000 sec / 0.000 sec
31 15:36:18 CREATE VIEW Account_View AS SELECT Account_No, Account_Type, Balance, Branch FROM ... 0 rows(s) affected 0.031 sec
32 15:36:28 SELECT * FROM Account_View LIMIT 0, 1000 10 rows(s) returned 0.000 sec / 0.000 sec
33 15:36:56 CREATE VIEW Savings_Account_View AS SELECT * FROM Account WHERE Account_Type = 'Savings' 0 rows(s) affected 0.015 sec
34 15:37:08 SELECT * FROM Savings_Account_View LIMIT 0, 1000 6 rows(s) returned 0.000 sec / 0.000 sec
35 15:37:52 CREATE VIEW Current_Account_View AS SELECT * FROM Account WHERE Account_Type = 'Current' 0 rows(s) affected 0.031 sec
36 15:38:01 SELECT * FROM Current_Account_View LIMIT 0, 1000 4 rows(s) returned 0.000 sec / 0.000 sec
```

16. HIGH BALANCE ACCOUNT VIEW

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following SQL script:

```
177 Account_No,
178 Customer_ID,
179 Account_Type,
180 Balance
181 FROM Account
182 WHERE Balance > 100000;
183 SELECT * FROM High_Balance_View;
```

The Results tab displays the output of the query:

Account_No	Customer_ID	Account_Type	Balance
10001	101	Current	120000.00
10007	107	Current	130000.00
10010	110	Current	120000.00

The Output tab shows the execution log:

#	Time	Action	Message	Duration / Fetch
28	15:34:52	SELECT * FROM Customer_View LIMIT 0, 1000	10 rows returned	0.000 sec / 0.000 sec
29	15:35:35	CREATE VIEW Customer_Basic_View AS SELECT Customer_ID, Customer_Name, City FROM Customer	0 rows affected	0.047 sec
30	15:35:45	SELECT * FROM Customer_Basic_View LIMIT 0, 1000	10 rows returned	0.000 sec / 0.000 sec
31	15:36:18	CREATE VIEW Account_View AS SELECT Account_No, Account_Type, Balance, Branch FROM	0 rows affected	0.031 sec
32	15:36:28	SELECT * FROM Account_View LIMIT 0, 1000	10 rows returned	0.000 sec / 0.000 sec
33	15:36:56	CREATE VIEW Savings_Account_View AS SELECT FROM Account WHERE Account_Type = 'Savings'	0 rows affected	0.015 sec
34	15:37:08	SELECT * FROM Savings_Account_View LIMIT 0, 1000	6 rows returned	0.000 sec / 0.000 sec
35	15:37:52	CREATE VIEW Current_Account_View AS SELECT FROM Account WHERE Account_Type = 'Current'	0 rows affected	0.031 sec
36	15:38:01	SELECT * FROM Current_Account_View LIMIT 0, 1000	4 rows returned	0.000 sec / 0.000 sec
37	15:38:31	CREATE VIEW High_Balance_View AS SELECT Account_No, Customer_ID, Account_Type, Balan...	0 rows affected	0.016 sec
38	15:38:40	SELECT * FROM High_Balance_View LIMIT 0, 1000	3 rows returned	0.000 sec / 0.000 sec

17. HYDERABAD ACCOUNT VIEW

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following SQL script:

```
183 SELECT * FROM High_Balance_View;
184
185 CREATE VIEW Hyderabad_Account_View AS
186 SELECT *
187 FROM Account
188 WHERE Branch = 'Hyderabad';
189 SELECT * FROM Hyderabad_Account_View;
```

The Results tab displays the output of the query:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10005	105	Current	90000.00	Hyderabad
10009	109	Savings	85000.00	Hyderabad

The Output tab shows the execution log:

#	Time	Action	Message	Duration / Fetch
30	15:35:45	SELECT * FROM Customer_Basic_View LIMIT 0, 1000	10 rows returned	0.000 sec / 0.000 sec
31	15:36:18	CREATE VIEW Account_View AS SELECT Account_No, Account_Type, Balance, Branch FROM	0 rows affected	0.031 sec
32	15:36:28	SELECT * FROM Account_View LIMIT 0, 1000	10 rows returned	0.000 sec / 0.000 sec
33	15:36:56	CREATE VIEW Savings_Account_View AS SELECT FROM Account WHERE Account_Type = 'Savings'	0 rows affected	0.015 sec
34	15:37:08	SELECT * FROM Savings_Account_View LIMIT 0, 1000	6 rows returned	0.000 sec / 0.000 sec
35	15:37:52	CREATE VIEW Current_Account_View AS SELECT FROM Account WHERE Account_Type = 'Current'	0 rows affected	0.031 sec
36	15:38:01	SELECT * FROM Current_Account_View LIMIT 0, 1000	4 rows returned	0.000 sec / 0.000 sec
37	15:38:31	CREATE VIEW High_Balance_View AS SELECT Account_No, Customer_ID, Account_Type, Balan...	0 rows affected	0.016 sec
38	15:38:40	SELECT * FROM High_Balance_View LIMIT 0, 1000	3 rows returned	0.000 sec / 0.000 sec
39	15:39:07	CREATE VIEW Hyderabad_Account_View AS SELECT FROM Account WHERE Branch = 'Hyderabad'	0 rows affected	0.016 sec
40	15:39:16	SELECT * FROM Hyderabad_Account_View LIMIT 0, 1000	3 rows returned	0.000 sec / 0.000 sec

18. LOW BALANCE ACCOUNT VIEW

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following SQL script:

```
192 SELECT
193 Account_No,
194 Customer_ID,
195 Balance
196 FROM Account
197 WHERE Balance < 50000;
198 SELECT * FROM Low_Balance_View;
```

The Results tab displays the output of the query:

Account_No	Customer_ID	Balance
10004	104	40000.00
10008	108	30000.00

The Output tab shows the execution log:

#	Time	Action	Message	Duration / Fetch
32	15:36:28	SELECT * FROM Account_View LIMIT 0, 1000	10 rows returned	0.000 sec / 0.000 sec
33	15:36:56	CREATE VIEW Savings_Account_View AS SELECT FROM Account WHERE Account_Type = 'Savings'	0 rows affected	0.015 sec
34	15:37:08	SELECT * FROM Savings_Account_View LIMIT 0, 1000	6 rows returned	0.000 sec / 0.000 sec
35	15:37:52	CREATE VIEW Current_Account_View AS SELECT FROM Account WHERE Account_Type = 'Current'	0 rows affected	0.031 sec
36	15:38:01	SELECT * FROM Current_Account_View LIMIT 0, 1000	4 rows returned	0.000 sec / 0.000 sec
37	15:38:31	CREATE VIEW High_Balance_View AS SELECT Account_No, Customer_ID, Account_Type, Balan...	0 rows affected	0.016 sec
38	15:38:40	SELECT * FROM High_Balance_View LIMIT 0, 1000	3 rows returned	0.000 sec / 0.000 sec
39	15:39:07	CREATE VIEW Hyderabad_Account_View AS SELECT FROM Account WHERE Branch = 'Hyderabad'	0 rows affected	0.016 sec
40	15:39:16	SELECT * FROM Hyderabad_Account_View LIMIT 0, 1000	3 rows returned	0.000 sec / 0.000 sec
41	15:39:43	CREATE VIEW Low_Balance_View AS SELECT Account_No, Customer_ID, Balance FROM Account...	0 rows affected	0.016 sec
42	15:39:54	SELECT * FROM Low_Balance_View LIMIT 0, 1000	2 rows returned	0.000 sec / 0.000 sec

19. CUSTOMER + ACCOUNT VIEW

MySQL Workbench

252009181_hmasree

File Edit View Query Database Server Tools Scripting Help

Navigator

Schemas

Filter objects

252009181_hmasree

Tables

Views

Stored Procedures

Functions

252009181_hmasree_pract

Tables

Views

Stored Procedures

Functions

252009181_hmasree_skillz

Tables

Views

Stored Procedures

Functions

252009181_hmasree_sq

Tables

Administration Schemas

Information

No object selected

Object Info Session

SQL Editor

219 A.Account_Type,

220 A.Balance,

221 A.Branch

222 FROM Customer C

223 JOIN Account A

224 ON C.Customer_ID = A.Customer_ID

225 SELECT * FROM Customer_Account_View

Results Grid

Filter Rows

Export

Wrap Cell Contents

Customer_ID Customer_Name City Account_No Account_Type Balance Branch

101 Ravi Kumar Hyderabad 10001 Savings 50000.00 Hyderabad

102 Priya Sharma Vijayawada 10002 Savings 75000.00 Vijayawada

103 Arjun Reddy Bangalore 10003 Current 120000.00 Bangalore

104 Sneha Rao Chennai 10004 Savings 45000.00 Chennai

105 Kiran Kumar Hyderabad 10005 Current 60000.00 Hyderabad

106 Anil Kumar Delhi 10006 Savings 65000.00 Delhi

107 Meera Reddy Mumbai 10007 Current 150000.00 Mumbai

108 Rahul Sharma Pune 10008 Savings 35000.00 Pune

Output

Action Output

Time Action Message Duration / Fetch

34 15:37:08 SELECT * FROM Savings_Account_View LIMIT 0, 1000 6 rows returned 0.000 sec / 0.000 sec

35 15:37:52 CREATE VIEW Current_Account_View AS SELECT * FROM Account WHERE Account_Type = 'Current' 0 rows affected 0.031 sec

36 15:38:01 SELECT * FROM Current_Account_View LIMIT 0, 1000 4 rows returned 0.000 sec / 0.000 sec

37 15:38:31 CREATE VIEW High_Balance_View AS SELECT Account_No, Customer_ID, Account_Type, Balan... 0 rows affected 0.016 sec

38 15:38:40 SELECT * FROM High_Balance_View LIMIT 0, 1000 3 rows returned 0.000 sec / 0.000 sec

39 15:39:07 CREATE VIEW Hyderabad_Account_View AS SELECT * FROM Account WHERE Branch = 'Hyderabad' 0 rows affected 0.016 sec

40 15:39:16 SELECT * FROM Hyderabad_Account_View LIMIT 0, 1000 3 rows returned 0.000 sec / 0.000 sec

41 15:39:43 CREATE VIEW Low_Balance_View AS SELECT Account_No, Customer_ID, Balance FROM Account... 0 rows affected 0.016 sec

42 15:39:54 SELECT * FROM Low_Balance_View LIMIT 0, 1000 2 rows returned 0.000 sec / 0.000 sec

43 15:40:25 CREATE VIEW Customer_Account_View AS SELECT C.Customer_ID, C.Customer_Name, C.City, A... 0 rows affected 0.016 sec

44 15:40:35 SELECT * FROM Customer_Account_View LIMIT 0, 1000 10 rows returned 0.000 sec / 0.000 sec

20. HYDERABAD CUSTOMER ACCOUNT VIEW

MySQL Workbench

252009181_hmasree

File Edit View Query Database Server Tools Scripting Help

Navigator

Schemas

Filter objects

252009181_hmasree

Tables

Views

Stored Procedures

Functions

252009181_hmasree_pract

Tables

Views

Stored Procedures

Functions

252009181_hmasree_skillz

Tables

Views

Stored Procedures

Functions

252009181_hmasree_sq

Tables

Administration Schemas

Information

No object selected

Object Info Session

SQL Editor

219 A.Account_Type,

220 A.Balance,

221 A.Branch

222 FROM Customer C

223 JOIN Account A

224 ON C.Customer_ID = A.Customer_ID

225 WHERE A.Branch = 'Hyderabad'

226 SELECT * FROM Hyderabad_Customer_Accounts

Results Grid

Filter Rows

Export

Wrap Cell Contents

Customer_Name City Account_No Account_Type Balance

Ravi Kumar Hyderabad 10001 Savings 50000.00

Kiran Kumar Hyderabad 10005 Current 60000.00

Lakshmi Devi Hyderabad 10009 Savings 85000.00

Hyderabad_Customer_Accounts

Output

Action Output

Time Action Message Duration / Fetch

36 15:38:01 SELECT * FROM Current_Account_View LIMIT 0, 1000 4 rows returned 0.000 sec / 0.000 sec

37 15:38:31 CREATE VIEW High_Balance_View AS SELECT Account_No, Customer_ID, Account_Type, Balan... 0 rows affected 0.016 sec

38 15:38:40 SELECT * FROM High_Balance_View LIMIT 0, 1000 3 rows returned 0.000 sec / 0.000 sec

39 15:39:07 CREATE VIEW Hyderabad_Account_View AS SELECT * FROM Account WHERE Branch = 'Hyderabad' 0 rows affected 0.016 sec

40 15:39:16 SELECT * FROM Hyderabad_Account_View LIMIT 0, 1000 3 rows returned 0.000 sec / 0.000 sec

41 15:39:43 CREATE VIEW Low_Balance_View AS SELECT Account_No, Customer_ID, Balance FROM Account... 0 rows affected 0.016 sec

42 15:39:54 SELECT * FROM Low_Balance_View LIMIT 0, 1000 2 rows returned 0.000 sec / 0.000 sec

43 15:40:25 CREATE VIEW Customer_Account_View AS SELECT C.Customer_ID, C.Customer_Name, C.City, A... 0 rows affected 0.016 sec

44 15:40:35 SELECT * FROM Customer_Account_View LIMIT 0, 1000 10 rows returned 0.000 sec / 0.000 sec

45 15:41:07 CREATE VIEW Hyderabad_Customer_Accounts AS SELECT C.Customer_Name, C.City, A.Account_N... 0 rows affected 0.016 sec

46 15:41:18 SELECT * FROM Hyderabad_Customer_Accounts LIMIT 0, 1000 3 rows returned 0.000 sec / 0.000 sec

21. CUSTOMER LOAN VIEW

MySQL Workbench

252009181_hmasree

File Edit View Query Database Server Tools Scripting Help

Navigator

Schemas

Filter objects

252009181_hmasree

Tables

Views

Stored Procedures

Functions

252009181_hmasree_pract

Tables

Views

Stored Procedures

Functions

252009181_hmasree_skillz

Tables

Views

Stored Procedures

Functions

252009181_hmasree_sq

Tables

Administration Schemas

Information

No object selected

Object Info Session

SQL Editor

233 L.Loan_Type,

234 L.Loan_Amount,

235 L.Interest_Rate

236 FROM Customer C

237 JOIN Loan L

238 ON C.Customer_ID = L.Customer_ID

239 SELECT * FROM Customer_Loan_View

Results Grid

Filter Rows

Export

Wrap Cell Contents

Customer_ID Customer_Name City Loan_ID Loan_Type Loan_Amount Interest_Rate

101 Ravi Kumar Hyderabad 101 Home Loan 500000.00 7.50

102 Priya Sharma Vijayawada 102 Education Loan 100000.00 6.50

103 Arjun Reddy Bangalore 103 Car Loan 80000.00 8.20

104 Sneha Rao Chennai 104 Personal Loan 30000.00 10.50

105 Kiran Kumar Hyderabad 105 Home Loan 400000.00 7.20

106 Anil Kumar Delhi 106 Car Loan 90000.00 8.50

107 Meera Reddy Mumbai 107 Business Loan 200000.00 9.80

108 Lakshmi Devi Hyderabad 108 Personal Loan 60000.00 10.00

Output

Action Output

Time Action Message Duration / Fetch

38 15:38:40 SELECT * FROM High_Balance_View LIMIT 0, 1000 3 rows returned 0.000 sec / 0.000 sec

39 15:39:07 CREATE VIEW Hyderabad_Account_View AS SELECT * FROM Account WHERE Branch = 'Hyderabad' 0 rows affected 0.016 sec

40 15:39:16 SELECT * FROM Hyderabad_Account_View LIMIT 0, 1000 3 rows returned 0.000 sec / 0.000 sec

41 15:39:43 CREATE VIEW Low_Balance_View AS SELECT Account_No, Customer_ID, Balance FROM Account... 0 rows affected 0.016 sec

42 15:39:54 SELECT * FROM Low_Balance_View LIMIT 0, 1000 2 rows returned 0.000 sec / 0.000 sec

43 15:40:25 CREATE VIEW Customer_Account_View AS SELECT C.Customer_ID, C.Customer_Name, C.City, A... 0 rows affected 0.016 sec

44 15:40:35 SELECT * FROM Customer_Account_View LIMIT 0, 1000 10 rows returned 0.000 sec / 0.000 sec

45 15:41:07 CREATE VIEW Hyderabad_Customer_Accounts AS SELECT C.Customer_Name, C.City, A.Account_N... 0 rows affected 0.016 sec

46 15:41:18 SELECT * FROM Hyderabad_Customer_Accounts LIMIT 0, 1000 3 rows returned 0.000 sec / 0.000 sec

47 15:41:55 CREATE VIEW Customer_Loan_View AS SELECT C.Customer_ID, C.Customer_Name, C.City, L.Lo... 0 rows affected 0.015 sec

48 15:42:04 SELECT * FROM Customer_Loan_View LIMIT 0, 1000 8 rows returned 0.000 sec / 0.000 sec

22. COMPLETE CUSTOMER BANKING VIEW

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```
253 L.Loan_Amount
254 FROM Customer C
255 LEFT JOIN Account A
256 ON C.Customer_ID = A.Customer_ID
257 LEFT JOIN Loan L
258 ON C.Customer_ID = L.Customer_ID
259 SELECT * FROM Customer_Banking_View
```

Customer_ID	Customer_Name	City	Account_No	Account_Type	Balance	Branch	Loan_Type	Loan_Amount
991	Ravi Kumar	Hyderabad	20001	Savings	50000.00	Hyderabad	Home Loan	500000.00
992	Priya Sharma	Vijayawada	20002	Savings	70000.00	Vijayawada	Education Loan	100000.00
993	Arun Reddy	Bangalore	20003	Current	120000.00	Bangalore	Car Loan	800000.00
994	Rohini Rao	Chennai	20004	Savings	40000.00	Chennai	Personal Loan	60000.00
995	Kiran Kumar	Hyderabad	20005	Current	90000.00	Hyderabad	Home Loan	400000.00
996	Aad Kumar	Delhi	20006	Savings	60000.00	Delhi	Car Loan	900000.00
997	Renee Reddy	Mumbai	20007	Current	100000.00	Mumbai	Business Loan	300000.00
998	Rahul Sharma	Pune	20008	Savings	30000.00	Pune	Home Loan	200000.00

Output

#	Time	Action	Message	Duration / Fetch
40	15:39:16	SELECT * FROM Hyderabad_Account_View LIMIT 5: 1000	3 row(s) returned	0.000 sec / 0.000 sec
41	15:39:43	CREATE VIEW Low_Balance_View AS SELECT Account_No, Customer_ID, Balance FROM Account	0 row(s) affected	0.016 sec
42	15:39:54	SELECT * FROM Low_Balance_View LIMIT 5: 1000	2 row(s) returned	0.000 sec / 0.000 sec
43	15:40:25	CREATE VIEW Customer_Account_View AS SELECT C.Customer_ID, C.Customer_Name, C.City, A.	0 row(s) affected	0.016 sec
44	15:40:35	SELECT * FROM Customer_Account_View LIMIT 0: 1000	10 row(s) returned	0.000 sec / 0.000 sec
45	15:41:07	CREATE VIEW Hyderabad_Customer_Accounts AS SELECT C.Customer_Name, C.City, A.Account_N.	0 row(s) affected	0.016 sec
46	15:41:18	SELECT * FROM Hyderabad_Customer_Accounts LIMIT 0: 1000	3 row(s) returned	0.000 sec / 0.000 sec
47	15:41:55	CREATE VIEW Customer_Loan_View AS SELECT C.Customer_ID, C.Customer_Name, C.City, L.Lo.	0 row(s) affected	0.015 sec
48	15:42:04	SELECT * FROM Customer_Loan_View LIMIT 0: 1000	8 row(s) returned	0.000 sec / 0.000 sec
49	15:42:41	CREATE VIEW Customer_Banking_View AS SELECT C.Customer_ID, C.Customer_Name, C.City, A.	0 row(s) affected	0.015 sec
50	15:42:51	SELECT * FROM Customer_Banking_View LIMIT 0: 1000	10 row(s) returned	0.000 sec / 0.000 sec
51	15:43:21	CREATE VIEW Total_Bank_Balance AS SELECT SUM(Balance) AS Total_Balance FROM Account	0 row(s) affected	0.016 sec
52	15:43:30	SELECT * FROM Total_Bank_Balance LIMIT 0: 1000	1 row(s) returned	0.016 sec / 0.000 sec

23. TOTAL BANK BALANCE

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```
258
259
260 CREATE VIEW Total_Bank_Balance AS
261 SELECT
262 SUM(Balance) AS Total_Balance
263 FROM Account
264 SELECT * FROM Total_Bank_Balance
```

Total_Balance
823000.00

Output

#	Time	Action	Message	Duration / Fetch
42	15:39:44	SELECT * FROM Low_Balance_View LIMIT 0: 1000	2 row(s) returned	0.000 sec / 0.000 sec
43	15:40:25	CREATE VIEW Customer_Account_View AS SELECT C.Customer_ID, C.Customer_Name, C.City, A.	0 row(s) affected	0.016 sec
44	15:40:35	SELECT * FROM Customer_Account_View LIMIT 0: 1000	10 row(s) returned	0.000 sec / 0.000 sec
45	15:41:07	CREATE VIEW Hyderabad_Customer_Accounts AS SELECT C.Customer_Name, C.City, A.Account_N.	0 row(s) affected	0.016 sec
46	15:41:18	SELECT * FROM Hyderabad_Customer_Accounts LIMIT 0: 1000	3 row(s) returned	0.000 sec / 0.000 sec
47	15:41:55	CREATE VIEW Customer_Loan_View AS SELECT C.Customer_ID, C.Customer_Name, C.City, L.Lo.	0 row(s) affected	0.015 sec
48	15:42:04	SELECT * FROM Customer_Loan_View LIMIT 0: 1000	8 row(s) returned	0.000 sec / 0.000 sec
49	15:42:41	CREATE VIEW Customer_Banking_View AS SELECT C.Customer_ID, C.Customer_Name, C.City, A.	0 row(s) affected	0.015 sec
50	15:42:51	SELECT * FROM Customer_Banking_View LIMIT 0: 1000	10 row(s) returned	0.000 sec / 0.000 sec
51	15:43:21	CREATE VIEW Total_Bank_Balance AS SELECT SUM(Balance) AS Total_Balance FROM Account	0 row(s) affected	0.016 sec
52	15:43:30	SELECT * FROM Total_Bank_Balance LIMIT 0: 1000	1 row(s) returned	0.016 sec / 0.000 sec

24. AVERAGE ACCOUNT BALANCE

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```
265
266 CREATE VIEW Average_Account_Balance AS
267 SELECT
268 AVG(Balance) AS Average_Balance
269 FROM Account
270
271 SELECT * FROM Average_Account_Balance
```

Average_Balance
82300.000000

Output

#	Time	Action	Message	Duration / Fetch
45	15:41:07	CREATE VIEW Hyderabad_Customer_Accounts AS SELECT C.Customer_Name, C.City, A.Account_N.	0 row(s) affected	0.016 sec
46	15:41:18	SELECT * FROM Hyderabad_Customer_Accounts LIMIT 0: 1000	3 row(s) returned	0.000 sec / 0.000 sec
47	15:41:55	CREATE VIEW Customer_Loan_View AS SELECT C.Customer_ID, C.Customer_Name, C.City, L.Lo.	0 row(s) affected	0.015 sec
48	15:42:04	SELECT * FROM Customer_Loan_View LIMIT 0: 1000	8 row(s) returned	0.000 sec / 0.000 sec
49	15:42:41	CREATE VIEW Customer_Banking_View AS SELECT C.Customer_ID, C.Customer_Name, C.City, A.	0 row(s) affected	0.015 sec
50	15:42:51	SELECT * FROM Customer_Banking_View LIMIT 0: 1000	10 row(s) returned	0.000 sec / 0.000 sec
51	15:43:21	CREATE VIEW Total_Bank_Balance AS SELECT SUM(Balance) AS Total_Balance FROM Account	0 row(s) affected	0.016 sec
52	15:43:30	SELECT * FROM Total_Bank_Balance LIMIT 0: 1000	1 row(s) returned	0.016 sec / 0.000 sec
53	15:44:09	CREATE VIEW Average_Account_Balance AS SELECT AVG(Balance) AS Average_Balance FROM Acco.	0 row(s) affected	0.031 sec
54	15:45:26	CREATE VIEW Average_Account_Balance AS SELECT AVG(Balance) AS Average_Balance FROM Acco.	Error Code: 1058. Table 'Average_Account_Balance' already exists	0.000 sec
55	15:45:29	SELECT * FROM Average_Account_Balance LIMIT 0: 1000	1 row(s) returned	0.015 sec / 0.000 sec

25. MAXIMUM ACCOUNT BALANCE

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```
271 SELECT * FROM Average_Account_Balance;
272
273 CREATE VIEW Maximum_Balance_View AS
274 SELECT
275 MAX(Balance) AS Maximum_Balance
276 FROM Account;
277 SELECT * FROM Maximum_Balance_View;
```

The Results grid shows the output of the last query:

Maximum_Balance
150000.00

The Output tab shows the execution log:

#	Time	Action	Message	Duration / Fetch
47	15:41:55	CREATE VIEW Customer_Loan_View AS SELECT C.Customer_ID, C.Customer_Name, C.City, L.Loan_Amount	0 row(s) affected	0.015 sec
48	15:42:04	SELECT * FROM Customer_Loan_View LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
49	15:42:41	CREATE VIEW Customer_Banking_View AS SELECT C.Customer_ID, C.Customer_Name, C.City, A.Account	0 row(s) affected	0.015 sec
50	15:42:51	SELECT * FROM Customer_Banking_View LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
51	15:43:21	CREATE VIEW Total_Bank_Balance AS SELECT SUM(Balance) AS Total_Balance FROM Account	0 row(s) affected	0.016 sec
52	15:43:30	SELECT * FROM Total_Bank_Balance LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec
53	15:44:09	CREATE VIEW Average_Account_Balance AS SELECT AVG(Balance) AS Average_Balance FROM Account	0 row(s) affected	0.031 sec
54	15:45:26	CREATE VIEW Average_Account_Balance AS SELECT AVG(Balance) AS Average_Balance FROM Account	Error Code: 1050. Table 'Average_Account_Balance' already exists	0.000 sec
55	15:45:29	SELECT * FROM Average_Account_Balance LIMIT 0, 1000	1 row(s) returned	0.015 sec / 0.000 sec
56	15:45:57	CREATE VIEW Maximum_Balance_View AS SELECT MAX(Balance) AS Maximum_Balance FROM Account	0 row(s) affected	0.015 sec
57	15:46:06	SELECT * FROM Maximum_Balance_View LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec

26. MINIMUM ACCOUNT BALANCE

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```
277 SELECT * FROM Maximum_Balance_View;
278
279 CREATE VIEW Minimum_Balance_View AS
280 SELECT
281 MIN(Balance) AS Minimum_Balance
282 FROM Account;
283 SELECT * FROM Minimum_Balance_View;
```

The Results grid shows the output of the last query:

Minimum_Balance
35000.00

The Output tab shows the execution log:

#	Time	Action	Message	Duration / Fetch
1	17:46:31	CREATE VIEW Minimum_Balance_View AS SELECT MIN(Balance) AS Minimum_Balance FROM Account	Error Code: 1046. No database selected. Select the default DB to be used by double-clicking its name in the SCH...	0.000 sec
2	17:46:36	USE 2520090181_himasee	0 row(s) affected	0.000 sec
3	17:46:39	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL;	Error Code: 1050. Table 'customer' already exists	0.031 sec
4	17:46:39	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'S,	Error Code: 1062. Duplicate entry '10001' for key 'account PRIMARY'	0.047 sec
5	17:46:54	CREATE VIEW Minimum_Balance_View AS SELECT MIN(Balance) AS Minimum_Balance FROM Account	0 row(s) affected	0.031 sec
6	17:47:08	SELECT * FROM Minimum_Balance_View LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

27. TOTAL NUMBER OF ACCOUNTS

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```
284
285
286 CREATE VIEW Account_Count_View AS
287 SELECT
288 COUNT(*) AS Total_Accounts
289 FROM Account;
290 SELECT * FROM Account_Count_View;
```

The Results grid shows the output of the last query:

Total_Accounts
10

The Output tab shows the execution log:

#	Time	Action	Message	Duration / Fetch
1	17:46:31	CREATE VIEW Minimum_Balance_View AS SELECT MIN(Balance) AS Minimum_Balance FROM Account	Error Code: 1046. No database selected. Select the default DB to be used by double-clicking its name in the SCH...	0.000 sec
2	17:46:36	USE 2520090181_himasee	0 row(s) affected	0.000 sec
3	17:46:39	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL;	Error Code: 1050. Table 'customer' already exists	0.031 sec
4	17:46:39	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'S,	Error Code: 1062. Duplicate entry '10001' for key 'account PRIMARY'	0.047 sec
5	17:46:54	CREATE VIEW Minimum_Balance_View AS SELECT MIN(Balance) AS Minimum_Balance FROM Account	0 row(s) affected	0.031 sec
6	17:47:08	SELECT * FROM Minimum_Balance_View LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
7	17:47:39	CREATE VIEW Account_Count_View AS SELECT COUNT(*) AS Total_Accounts FROM Account	0 row(s) affected	0.016 sec
8	17:47:44	SELECT * FROM Account_Count_View LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

28. BRANCH-WISE ACCOUNT COUNT

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following SQL code:

```
292 CREATE VIEW Branch_Account_Count AS
293 SELECT
294   Branch,
295   COUNT(*) AS Number_of_Accounts
296 FROM Account
297 GROUP BY Branch;
298 SELECT * FROM Branch_Account_Count;
```

The Result Grid displays the following data:

Branch	Number_of_Accounts
Hyderabad	3
Vijayawada	2
Bangalore	1
Chennai	1
Delhi	1
Mumbai	1
Pune	1

The Action Output pane shows the execution of the SQL statements:

#	Time	Action	Message	Duration / Fetch
1	21:59:21	CREATE VIEW Branch_Account_Count AS SELEC...	Error Code: 1046: No database selected Select the ...	0.000 sec
2	21:59:31	USE 2520090181_himasee	0 row(s) affected	0.000 sec
3	21:59:43	INSERT INTO Customer (Customer_ID, Customer_...	Error Code: 1062: Duplicate entry '101' for key 'cust...	0.016 sec
4	21:59:52	CREATE VIEW Branch_Account_Count AS SELEC...	0 row(s) affected	0.015 sec
5	22:00:02	SELECT * FROM Branch_Account_Count LIMIT 0 ...	7 row(s) returned	0.000 sec / 0.000 sec

29. BRANCH-WISE TOTAL BALANCE

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following SQL code:

```
300 CREATE VIEW Branch_Total_Balance AS
301 SELECT
302   Branch,
303   SUM(Balance) AS Total_Balance
304 FROM Account
305 GROUP BY Branch;
306 SELECT * FROM Branch_Total_Balance;
```

The Result Grid displays the following data:

Branch	Total_Balance
Hyderabad	225000.00
Vijayawada	185000.00
Bangalore	120000.00
Chennai	40000.00
Delhi	65000.00
Mumbai	150000.00
Pune	35000.00

The Action Output pane shows the execution of the SQL statements:

#	Time	Action	Message	Duration / Fetch
1	21:59:21	CREATE VIEW Branch_Account_Count AS SELEC...	Error Code: 1046: No database selected Select the ...	0.000 sec
2	21:59:31	USE 2520090181_himasee	0 row(s) affected	0.000 sec
3	21:59:43	INSERT INTO Customer (Customer_ID, Customer_...	Error Code: 1062: Duplicate entry '101' for key 'cust...	0.016 sec
4	21:59:52	CREATE VIEW Branch_Account_Count AS SELEC...	0 row(s) affected	0.015 sec
5	22:00:02	SELECT * FROM Branch_Account_Count LIMIT 0 ...	7 row(s) returned	0.000 sec / 0.000 sec
6	22:00:41	CREATE VIEW Branch_Total_Balance AS SELEC...	0 row(s) affected	0.000 sec
7	22:00:50	SELECT * FROM Branch_Total_Balance LIMIT 0 ...	7 row(s) returned	0.000 sec / 0.000 sec

30. ACCOUNT TYPE COUNT

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following SQL code:

```
309 CREATE VIEW Account_Type_Count AS
310 SELECT
311   Account_Type,
312   COUNT(*) AS Number_of_Accounts
313 FROM Account
314 GROUP BY Account_Type;
315 SELECT * FROM Account_Type_Count;
```

The Result Grid displays the following data:

Account_Type	Number_of_Accounts
Savings	6
Current	4

The Action Output pane shows the execution of the SQL statements:

#	Time	Action	Message	Duration / Fetch
1	21:59:21	CREATE VIEW Branch_Account_Count AS SELEC...	Error Code: 1046: No database selected Select the ...	0.000 sec
2	21:59:31	USE 2520090181_himasee	0 row(s) affected	0.000 sec
3	21:59:43	INSERT INTO Customer (Customer_ID, Customer_...	Error Code: 1062: Duplicate entry '101' for key 'cust...	0.016 sec
4	21:59:52	CREATE VIEW Branch_Account_Count AS SELEC...	0 row(s) affected	0.015 sec
5	22:00:02	SELECT * FROM Branch_Account_Count LIMIT 0 ...	7 row(s) returned	0.000 sec / 0.000 sec
6	22:00:41	CREATE VIEW Branch_Total_Balance AS SELEC...	0 row(s) affected	0.000 sec
7	22:00:50	SELECT * FROM Branch_Total_Balance LIMIT 0 ...	7 row(s) returned	0.000 sec / 0.000 sec
8	22:01:55	CREATE VIEW Account_Type_Count AS SELEC...	0 row(s) affected	0.016 sec
9	22:02:03	SELECT * FROM Account_Type_Count LIMIT 0.1 ...	2 row(s) returned	0.000 sec / 0.000 sec

31. ACCOUNT TYPE TOTAL BALANCE

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following code:

```
317 CREATE VIEW Account_Type_Balance AS
318 SELECT
319     Account_Type,
320     SUM(Balance) AS Total_Balance
321 FROM Account_Type
322 GROUP BY Account_Type;
```

The Results grid shows the output of the query:

Account_Type	Total_Balance
Savings	20000.00
Current	40000.00

The Action Output pane shows the execution log, including the creation of the view and the execution of the query.

32. BRANCHES WITH MORE THAN ONE ACCOUNT

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following code:

```
326 SELECT
327     Branch,
328     COUNT(*) AS Number_of_Accounts
329 FROM Account_Type
330 GROUP BY Branch
331 HAVING COUNT(*) > 1;
```

The Results grid shows the output of the query:

Branch	Number_of_Accounts
Hyderabad	3
Vijayawada	2

The Action Output pane shows the execution log, including the creation of the view and the execution of the query.

33. BRANCHES WITH TOTAL BALANCE ABOVE 100000

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following code:

```
335 SELECT
336     Branch,
337     SUM(Balance) AS Total_Balance
338 FROM Account_Type
339 GROUP BY Branch
340 HAVING SUM(Balance) > 100000;
```

The Results grid shows the output of the query:

Branch	Total_Balance
Hyderabad	225000.00
Vijayawada	85000.00
Bangalore	120000.00
Mumbai	150000.00

The Action Output pane shows the execution log, including the creation of the view and the execution of the query.

34. DEPOSIT TRANSACTION VIEW

The screenshot shows the MySQL Workbench interface with the 'Schemas' pane on the left displaying the '2520090181_himansree' database. The 'SQL Editor' pane contains the following SQL code:

```
341 * SELECT * FROM Rich_Branches;
342
343 CREATE VIEW Deposit_Transaction_View AS
344 SELECT *
345 FROM Bank_Transaction
346 WHERE Transaction_Type = 'DEPOSIT';
347 * SELECT * FROM Deposit_Transaction_View;
```

The 'Result Grid' pane shows the output of the final query, displaying a table with 10 rows of deposit transactions. The 'Output' pane shows the execution log, including the creation of the view.

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-09-12 15:22+
3	10002	DEPOSIT	15000.00	2026-09-12 15:22+
5	10004	DEPOSIT	5000.00	2026-09-12 15:22+
7	10006	DEPOSIT	12000.00	2026-09-12 15:22+
8	10007	DEPOSIT	25000.00	2026-09-12 15:22+
10	10009	DEPOSIT	20000.00	2026-09-12 15:22+

35. WITHDRAWAL TRANSACTION VIEW

The screenshot shows the MySQL Workbench interface with the 'Schemas' pane on the left displaying the '2520090181_himansree' database. The 'SQL Editor' pane contains the following SQL code:

```
347 * SELECT * FROM Deposit_Transaction_View;
348
349 CREATE VIEW Withdrawal_Transaction_View AS
350 SELECT *
351 FROM Bank_Transaction
352 WHERE Transaction_Type = 'WITHDRAW';
353 * SELECT * FROM Withdrawal_Transaction_View;
```

The 'Result Grid' pane shows the output of the final query, displaying a table with 11 rows of withdrawal transactions. The 'Output' pane shows the execution log, including the creation of the view.

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
2	10001	WITHDRAW	5000.00	2026-09-12 15:22+
4	10003	WITHDRAW	20000.00	2026-09-12 15:22+
6	10005	WITHDRAW	8000.00	2026-09-12 15:22+
9	10008	WITHDRAW	5000.00	2026-09-12 15:22+
11	10010	WITHDRAW	15000.00	2026-09-12 15:22+

36. TRANSACTION DETAILS WITH CUSTOMER

The screenshot shows the MySQL Workbench interface with the 'Schemas' pane on the left displaying the '2520090181_himansree' database. The 'SQL Editor' pane contains the following SQL code:

```
363 * T.Transaction_Date
364 FROM Customer C
365 JOIN Account A
366 ON C.Customer_ID = A.Customer_ID
367 JOIN Bank_Transaction T
368 ON A.Account_No = T.Account_No;
369 * SELECT * FROM Customer_Transaction_View;
```

The 'Result Grid' pane shows the output of the final query, displaying a table with 11 rows of transaction details for each customer. The 'Output' pane shows the execution log, including the creation of the view.

Customer_Name	Account_No	Account_Type	Transaction_ID	Transaction_Type
Ravi Kumar	10001	Savings	1	DEPOSIT
Ravi Kumar	10001	Savings	2	WITHDRAW
Prisha Sharma	10002	Savings	3	DEPOSIT
Akshu Baddi	10003	Current	4	WITHDRAW
Shreha Rao	10004	Savings	5	DEPOSIT
Kiran Kumar	10005	Current	6	WITHDRAW

37. HIGH VALUE TRANSACTION VIEW

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following code:

```
370
371 CREATE VIEW High_Value_Transaction_View AS
372 SELECT *
373 FROM Bank_Transaction
374 WHERE Amount > 10000;
375
376 SELECT * FROM High_Value_Transaction_View;
```

The Results grid shows the output of the SELECT statement:

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
3	10002	DEPOSIT	15000.00	2026-08-12 15:22+
4	10003	WITHDRAWAL	20000.00	2026-08-12 15:22+
7	10006	DEPOSIT	12000.00	2026-08-12 15:22+
8	10007	DEPOSIT	20000.00	2026-08-12 15:22+
10	10009	DEPOSIT	20000.00	2026-08-12 15:22+
11	10010	WITHDRAWAL	15000.00	2026-08-12 15:22+

The Action Output pane shows the execution log:

#	Time	Action	Message	Duration / Feat
13	22:03:54	SELECT * FROM Multiple_Account_Branches U...	2 row(s) returned	0.000 sec / 0.000 sec
14	22:04:19	CREATE VIEW Rich_Branches AS SELECT Br...	0 row(s) affected	0.016 sec
15	22:04:27	SELECT * FROM Rich_Branches LIMIT 0, 1000	4 row(s) returned	0.015 sec / 0.000 sec
16	22:05:39	CREATE VIEW Deposit_Transaction_View AS S...	0 row(s) affected	0.031 sec
17	22:05:46	SELECT * FROM Deposit_Transaction_View LIM...	6 row(s) returned	0.000 sec / 0.000 sec
18	22:06:24	CREATE VIEW Withdrawal_Transaction_View A...	0 row(s) affected	0.015 sec
19	22:06:34	SELECT * FROM Withdrawal_Transaction_View ...	5 row(s) returned	0.000 sec / 0.000 sec
20	22:07:19	CREATE VIEW Customer_Transaction_View AS ...	0 row(s) affected	0.016 sec
21	22:07:30	SELECT * FROM Customer_Transaction_View U...	11 row(s) returned	0.016 sec / 0.000 sec
22	22:08:24	CREATE VIEW High_Value_Transaction_View A...	0 row(s) affected	0.015 sec
23	22:08:34	SELECT * FROM High_Value_Transaction_View ...	6 row(s) returned	0.016 sec / 0.000 sec
24	22:08:58	CREATE VIEW Balance_Ranking_View AS SEL...	0 row(s) affected	0.015 sec
25	22:09:06	SELECT * FROM Balance_Ranking_View LIMIT ...	10 row(s) returned	0.000 sec / 0.000 sec

38. ACCOUNTS ORDERED BY BALANCE

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following code:

```
381 Account_No,
382 Customer_ID,
383 Account_Type,
384 Balance,
385 FROM Account
386 ORDER BY Balance DESC;
387 SELECT * FROM Balance_Ranking_View;
```

The Results grid shows the output of the SELECT statement:

Account_No	Customer_ID	Account_Type	Balance
10007	107	Current	150000.00
10003	103	Current	120000.00
10010	110	Current	110000.00
10009	109	Current	90000.00
10009	109	Savings	85000.00
10002	102	Savings	75000.00
10006	106	Savings	65000.00
10004	104	Fixed	60000.00

The Action Output pane shows the execution log:

#	Time	Action	Message	Duration / Feat
15	22:04:27	SELECT * FROM Rich_Branches LIMIT 0, 1000	4 row(s) returned	0.015 sec / 0.000 sec
16	22:05:39	CREATE VIEW Deposit_Transaction_View AS S...	0 row(s) affected	0.031 sec
17	22:05:46	SELECT * FROM Deposit_Transaction_View LIM...	6 row(s) returned	0.000 sec / 0.000 sec
18	22:06:24	CREATE VIEW Withdrawal_Transaction_View A...	0 row(s) affected	0.015 sec
19	22:06:34	SELECT * FROM Withdrawal_Transaction_View ...	5 row(s) returned	0.000 sec / 0.000 sec
20	22:07:19	CREATE VIEW Customer_Transaction_View AS ...	0 row(s) affected	0.016 sec
21	22:07:30	SELECT * FROM Customer_Transaction_View U...	11 row(s) returned	0.016 sec / 0.000 sec
22	22:08:24	CREATE VIEW High_Value_Transaction_View A...	0 row(s) affected	0.015 sec
23	22:08:34	SELECT * FROM High_Value_Transaction_View ...	6 row(s) returned	0.016 sec / 0.000 sec
24	22:08:58	CREATE VIEW Balance_Ranking_View AS SEL...	0 row(s) affected	0.015 sec
25	22:09:06	SELECT * FROM Balance_Ranking_View LIMIT ...	10 row(s) returned	0.000 sec / 0.000 sec

39. CUSTOMERS ORDERED BY NAME

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following code:

```
391 Customer_ID,
392 Customer_Name,
393 City,
394 FROM Customer
395 ORDER BY Customer_Name;
396
397 SELECT * FROM Customer_Name_View;
```

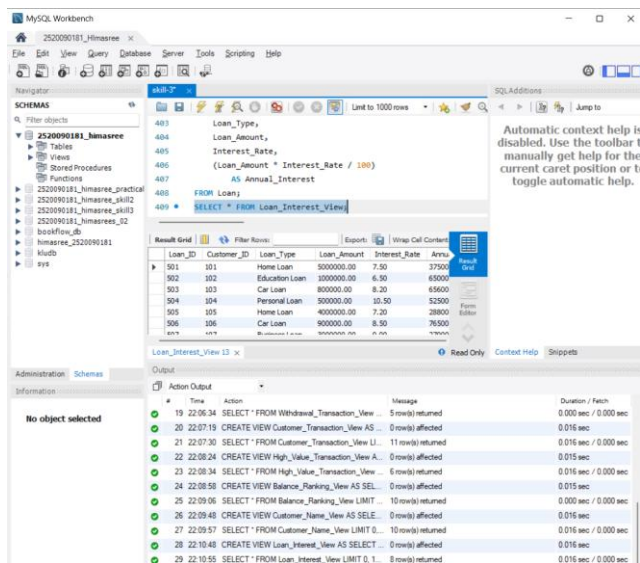
The Results grid shows the output of the SELECT statement:

Customer_ID	Customer_Name	City
106	Anil Kumar	Delhi
103	Arun Reddy	Bangalore
105	Vijay Kumar	Hyderabad
109	Lakshmi Devi	Hyderabad
107	Meena Reddy	Mumbai
102	Priya Sharma	Hyderabad
108	Rahul Sharma	Pune

The Action Output pane shows the execution log:

#	Time	Action	Message	Duration / Feat
17	22:05:48	SELECT * FROM Deposit_Transaction_View LIM...	6 row(s) returned	0.000 sec / 0.000 sec
18	22:06:34	CREATE VIEW Withdrawal_Transaction_View A...	0 row(s) affected	0.015 sec
19	22:06:34	SELECT * FROM Withdrawal_Transaction_View ...	5 row(s) returned	0.000 sec / 0.000 sec
20	22:07:19	CREATE VIEW Customer_Transaction_View AS ...	0 row(s) affected	0.016 sec
21	22:07:30	SELECT * FROM Customer_Transaction_View U...	11 row(s) returned	0.016 sec / 0.000 sec
22	22:08:24	CREATE VIEW High_Value_Transaction_View A...	0 row(s) affected	0.015 sec
23	22:08:34	SELECT * FROM High_Value_Transaction_View ...	6 row(s) returned	0.016 sec / 0.000 sec
24	22:08:58	CREATE VIEW Balance_Ranking_View AS SEL...	0 row(s) affected	0.015 sec
25	22:09:06	SELECT * FROM Balance_Ranking_View LIMIT ...	10 row(s) returned	0.000 sec / 0.000 sec
26	22:09:48	CREATE VIEW Customer_Name_View AS SEL...	0 row(s) affected	0.016 sec
27	22:09:57	SELECT * FROM Customer_Name_View LIMIT 0...	10 row(s) returned	0.016 sec / 0.000 sec

40. LOAN INTEREST CALCULATION VIEW



Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

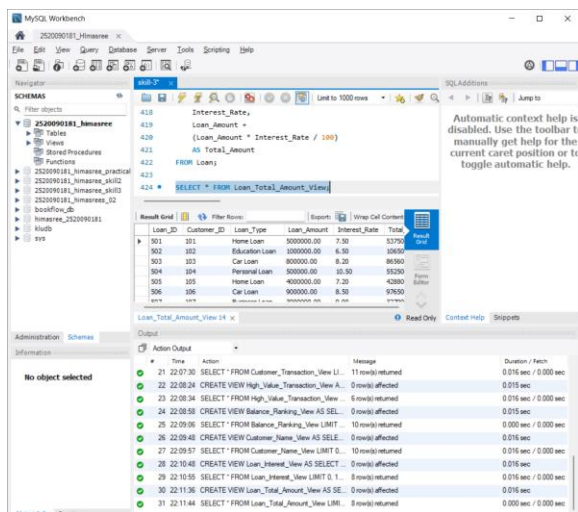
```
403 Loan_Type,  
404 Loan_Amount,  
405 Interest_Rate,  
406 (Loan_Amount * Interest_Rate / 100)  
407 AS Annual_Interest  
408 FROM Loan;  
409 CREATE VIEW Loan_Interest_View  
410 AS  
411 SELECT * FROM Loan;
```

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate	Annual_Interest
901	101	Home Loan	500000.00	7.50	37500
902	102	Education Loan	100000.00	6.50	6500
903	103	Car Loan	80000.00	8.20	6560
904	104	Personal Loan	50000.00	10.50	5250
905	105	Home Loan	400000.00	7.20	28800
906	106	Car Loan	90000.00	8.50	7650

Loan_Interest_View 13 x

#	Time	Action	Message	Duration / Fetch
19	22:06:34	SELECT * FROM Withdrawal_Transaction_View ...	5 row(s) returned	0.000 sec / 0.000 sec
20	22:07:19	CREATE VIEW Customer_Transaction_View AS ...	0 row(s) affected	0.016 sec
22	22:08:24	CREATE VIEW High_Value_Transaction_View AS ...	0 row(s) affected	0.016 sec / 0.000 sec
23	22:08:34	SELECT * FROM High_Value_Transaction_View ...	6 row(s) returned	0.016 sec / 0.000 sec
24	22:08:58	CREATE VIEW Balance_Ranking_View AS SELECT ...	0 row(s) affected	0.015 sec
25	22:09:06	SELECT * FROM Balance_Ranking_View LIMIT ...	10 row(s) returned	0.000 sec / 0.000 sec
26	22:09:48	CREATE VIEW Customer_Name_View AS SELECT ...	0 row(s) affected	0.016 sec
27	22:09:57	SELECT * FROM Customer_Name_View LIMIT 0 ...	10 row(s) returned	0.016 sec / 0.000 sec
28	22:10:48	CREATE VIEW Loan_Interest_View AS SELECT ...	0 row(s) affected	0.016 sec
29	22:10:55	SELECT * FROM Loan_Interest_View LIMIT 0.1 ...	8 row(s) returned	0.016 sec / 0.000 sec

41. LOAN TOTAL AMOUNT VIEW



Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

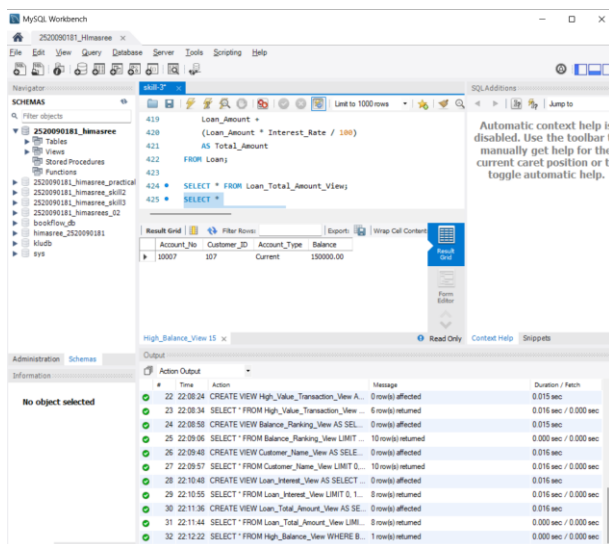
```
418 Interest_Rate,  
419 Loan_Amount +  
420 (Loan_Amount * Interest_Rate / 100)  
421 AS Total_Amount  
422 FROM Loan;  
423 CREATE VIEW Loan_Total_Amount_View  
424 AS  
425 SELECT * FROM Loan;
```

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate	Total_Amount
901	101	Home Loan	500000.00	7.50	53750
902	102	Education Loan	100000.00	6.50	10650
903	103	Car Loan	80000.00	8.20	86560
904	104	Personal Loan	50000.00	10.50	55250
905	105	Home Loan	400000.00	7.20	42880
906	106	Car Loan	90000.00	8.50	97650

Loan_Total_Amount_View 14 x

#	Time	Action	Message	Duration / Fetch
21	22:07:30	SELECT * FROM Customer_Transaction_View LI ...	11 row(s) returned	0.016 sec / 0.000 sec
22	22:08:24	CREATE VIEW High_Value_Transaction_View AS ...	0 row(s) affected	0.016 sec
23	22:08:34	SELECT * FROM High_Value_Transaction_View ...	6 row(s) returned	0.016 sec / 0.000 sec
24	22:08:58	CREATE VIEW Balance_Ranking_View AS SEL ...	0 row(s) affected	0.015 sec
25	22:09:06	SELECT * FROM Balance_Ranking_View LIMIT ...	10 row(s) returned	0.000 sec / 0.000 sec
26	22:09:48	CREATE VIEW Customer_Name_View AS SELE ...	0 row(s) affected	0.016 sec
27	22:09:57	SELECT * FROM Customer_Name_View LIMIT 0 ...	10 row(s) returned	0.016 sec / 0.000 sec
28	22:10:48	CREATE VIEW Loan_Interest_View AS SELECT ...	0 row(s) affected	0.016 sec
29	22:10:55	SELECT * FROM Loan_Interest_View LIMIT 0.1 ...	8 row(s) returned	0.016 sec / 0.000 sec
30	22:11:36	CREATE VIEW Loan_Total_Amount_View AS SE ...	0 row(s) affected	0.016 sec
31	22:11:44	SELECT * FROM Loan_Total_Amount_View LIM ...	8 row(s) returned	0.000 sec / 0.000 sec

42. DISPLAY HIGH BALANCE ACCOUNTS FROM VIEW



Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```
419 Loan_Amount +  
420 (Loan_Amount * Interest_Rate / 100)  
421 AS Total_Amount  
422 FROM Loan;  
423 CREATE VIEW Loan_Total_Amount_View  
424 AS  
425 SELECT * FROM Loan;  
426  
427 CREATE VIEW High_Balance_View  
428 AS  
429 SELECT * FROM Loan_Total_Amount_View  
430 WHERE Total_Amount > 100000.00;
```

Account_No	Customer_ID	Account_Type	Balance
10007	107	Current	150000.00

High_Balance_View 15 x

#	Time	Action	Message	Duration / Fetch
22	22:08:24	CREATE VIEW High_Value_Transaction_View AS ...	0 row(s) affected	0.015 sec
23	22:08:34	SELECT * FROM High_Value_Transaction_View ...	6 row(s) returned	0.016 sec / 0.000 sec
24	22:08:58	CREATE VIEW Balance_Ranking_View AS SELECT ...	0 row(s) affected	0.015 sec
25	22:09:06	SELECT * FROM Balance_Ranking_View LIMIT ...	10 row(s) returned	0.000 sec / 0.000 sec
26	22:09:48	CREATE VIEW Customer_Name_View AS SELE ...	0 row(s) affected	0.016 sec
27	22:09:57	SELECT * FROM Customer_Name_View LIMIT 0 ...	10 row(s) returned	0.016 sec / 0.000 sec
28	22:10:48	CREATE VIEW Loan_Interest_View AS SELECT ...	0 row(s) affected	0.016 sec
29	22:10:55	SELECT * FROM Loan_Interest_View LIMIT 0.1 ...	8 row(s) returned	0.016 sec / 0.000 sec
30	22:11:36	CREATE VIEW Loan_Total_Amount_View AS SE ...	0 row(s) affected	0.016 sec
31	22:11:44	SELECT * FROM Loan_Total_Amount_View LIM ...	8 row(s) returned	0.000 sec / 0.000 sec
32	22:12:22	SELECT * FROM High_Balance_View WHERE B ...	1 row(s) returned	0.000 sec / 0.000 sec

43. DISPLAY SAVINGS ACCOUNTS ABOVE 60000

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
SELECT * FROM Loan_Total_Amount_View  
SELECT *  
FROM High_Balance_View  
WHERE Balance > 120000  
SELECT *
```

The result grid displays the following data:

Account_No	Customer_ID	Account_Type	Balance	Branch
10002	102	Savings	75000.00	Vijayawada
10006	106	Savings	65000.00	Delhi
10009	109	Savings	85000.00	Hyderabad

The Action Output pane shows the execution of the query, indicating that 3 rows were returned.

44. DISPLAY HYDERABAD ACCOUNTS ABOVE 50000

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
FROM High_Balance_View  
WHERE Balance > 120000  
SELECT *  
FROM Savings_Account_View  
WHERE Balance > 60000
```

The result grid displays the following data:

Account_No	Customer_ID	Account_Type	Balance	Branch
10005	105	Current	90000.00	Hyderabad
10009	109	Savings	85000.00	Hyderabad

The Action Output pane shows the execution of the query, indicating that 2 rows were returned.

45. DISPLAY CUSTOMERS FROM HYDERABAD

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
WHERE Balance > 60000  
[Execute the selected portion of the script or everything, if there is no selection]  
SELECT *  
FROM Hyderabad_Account_View  
WHERE Balance > 50000  
SELECT *
```

The result grid displays the following data:

Customer_ID	Customer_Name	City
101	Ravi Kumar	Hyderabad
105	Kiran Kumar	Hyderabad
109	Lakshmi Devi	Hyderabad

The Action Output pane shows the execution of the query, indicating that 3 rows were returned.

46. DISPLAY CUSTOMER ACCOUNTS ORDERED BY BALANCE

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
435 WHERE Balance > 50000;
436
437 * SELECT *
438 FROM Customer_Basic_View
439 WHERE City = 'Hyderabad';
440
441 * SELECT *
```

The Results grid displays the following data:

Customer_ID	Customer_Name	City	Account_No	Account_Type	Balance
107	Meena Reddy	Mumbai	10007	Current	15000
101	Arun Reddy	Bangalore	10001	Current	12000
110	Suresh Babu	Vijayawada	10010	Current	11000
105	Kiran Kumar	Hyderabad	10005	Current	9000
109	Lakshmi Devi	Hyderabad	10009	Savings	8500
102	Priya Sharma	Vijayawada	10002	Savings	7500
106	Santhosh Kumar	Hyderabad	10006	Current	6000

The Action Output pane shows the execution of the query, indicating that 10 rows were returned.

47. DISPLAY LOANS ABOVE 1000000

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
441 * SELECT *
442 FROM Customer_Account_View
443 ORDER BY Balance DESC;
444
445 * SELECT *
```

The Results grid displays the following data:

Customer_ID	Customer_Name	City	Loan_ID	Loan_Type	Loan_Amount
101	Kiran Kumar	Hyderabad	101	Home Loan	3000000
105	Kiran Kumar	Hyderabad	105	Home Loan	4000000
107	Meena Reddy	Mumbai	107	Business Loan	3000000

The Action Output pane shows the execution of the query, indicating that 3 rows were returned.

48. UPDATE ACCOUNT BALANCE THROUGH VIEW

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
447 FROM Customer_Loan_View
448 WHERE Loan_Amount > 1000000;
449
450 * UPDATE Account_View
451 SET Balance = 10000;
452 WHERE Account_No = 10001;
453 * SELECT *
```

The Results grid displays the following data:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	60000.00	Hyderabad
10009	109	Savings	8500.00	Hyderabad

The Action Output pane shows the execution of the query, indicating that 1 row was affected.

49. INSERT USING SIMPLE VIEW

MySQL Workbench interface showing a query window with the following SQL code:

```
INSERT INTO Simple_Account_View  
VALUES  
(10011, 101, 'Savings', 55000, 'Hyderabad');
```

The query is executed, and the output shows the result grid with the following data:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	60000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
10006	106	Savings	65000.00	Delhi
10007	107	Current	150000.00	Mumbai

The output also shows the action output with the following message:

```
42 22:18:13 INSERT INTO Simple_Account_View VALUES (1... 1 row(s) affected  
43 22:18:43 SELECT * FROM Account LIMIT 0, 1000 11 row(s) returned
```

50. DELETE USING SIMPLE VIEW

MySQL Workbench interface showing a query window with the following SQL code:

```
DELETE FROM Simple_Account_View  
WHERE Account_No = 10011;
```

The query is executed, and the output shows the result grid with the following data:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	60000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
10006	106	Savings	65000.00	Delhi
10007	107	Current	150000.00	Mumbai

The output also shows the action output with the following message:

```
42 22:18:13 INSERT INTO Simple_Account_View VALUES (1... 1 row(s) affected  
43 22:18:43 SELECT * FROM Account LIMIT 0, 1000 11 row(s) returned  
44 22:19:13 DELETE FROM Simple_Account_View WHERE ... 1 row(s) affected  
45 22:19:19 SELECT * FROM Account LIMIT 0, 1000 10 row(s) returned
```

51. SHOW ALL VIEWS

MySQL Workbench interface showing a query window with the following SQL code:

```
SHOW FULL TABLES WHERE TABLE_TYPE = 'VIEW';
```

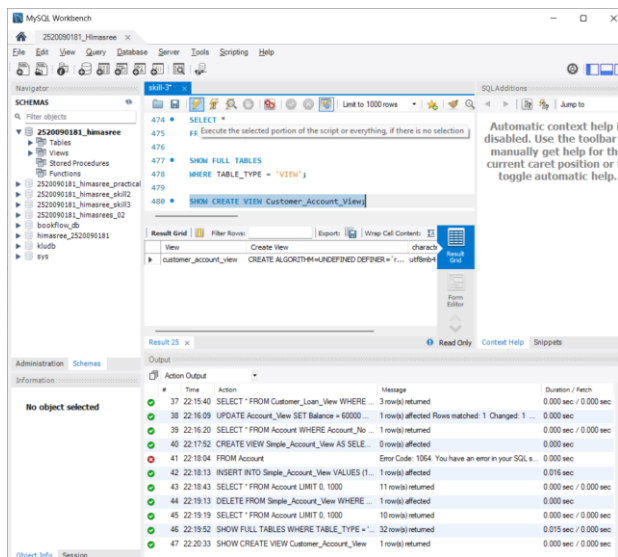
The query is executed, and the output shows the result grid with the following data:

Table_name	Table_type
account_count_view	VIEW
account_type_balance	VIEW
account_type_count	VIEW
account_view	VIEW
average_account_balance	VIEW
balance_saving_view	VIEW
branch_account_count	VIEW

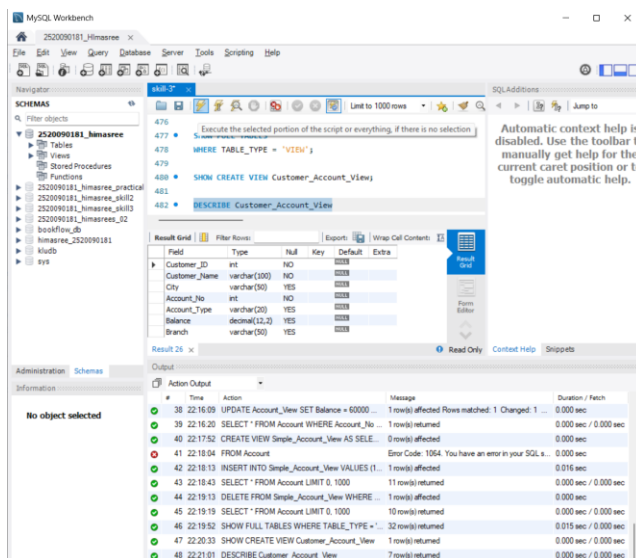
The output also shows the action output with the following message:

```
46 22:19:52 SHOW FULL TABLES WHERE TABLE_TYPE = 'VIEW' 32 row(s) returned
```

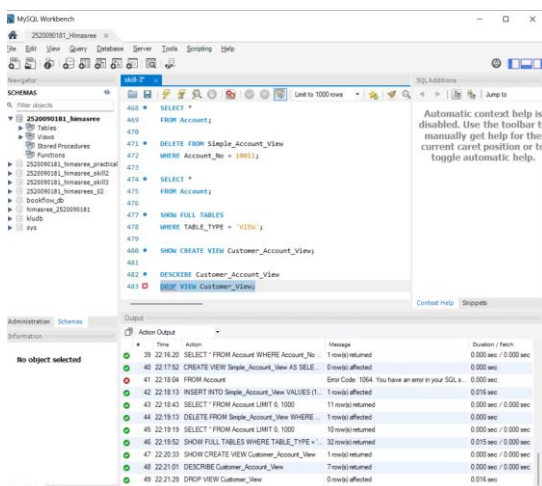
52. SHOW VIEW DEFINITION



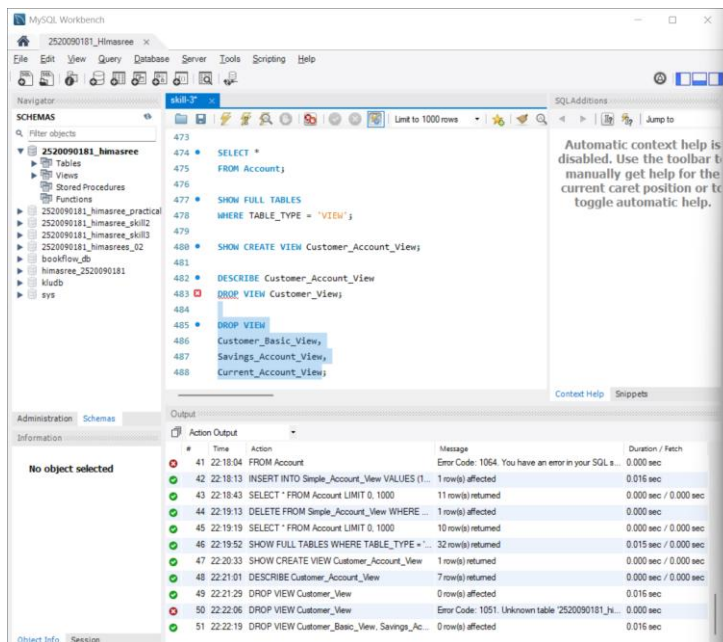
53. DESCRIBE VIEW



54. DROP ONE VIEW



55. DROP MULTIPLE VIEWS



The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

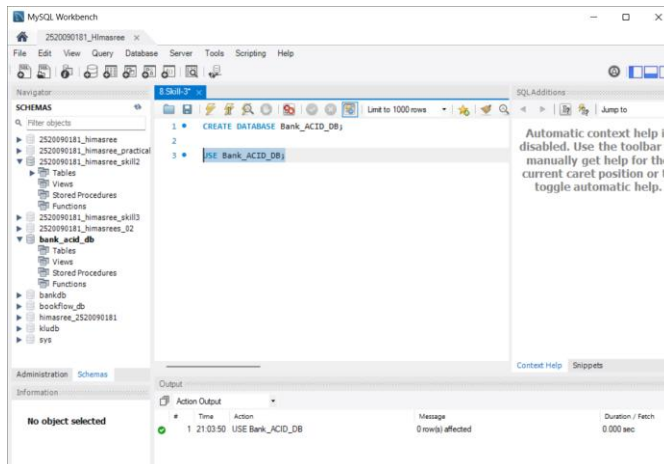
```
473
474 SELECT *
475 FROM Account;
476
477 SHOW FULL TABLES
478 WHERE TABLE_TYPE = 'VIEW';
479
480 SHOW CREATE VIEW Customer_Account_View;
481
482 DESCRIBE Customer_Account_View;
483 DROP VIEW Customer_View;
484
485 DROP VIEW
486 Customer_Basic_View,
487 Savings_Account_View,
488 Current_Account_View;
```

The output window shows the execution results:

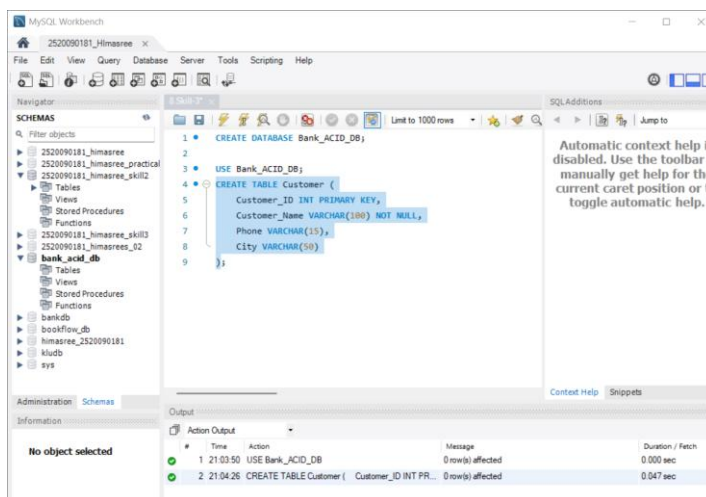
#	Time	Action	Message	Duration / Fetch
41	22:18:04	FROM Account	Error Code: 1054. You have an error in your SQL s...	0.000 sec
42	22:18:13	INSERT INTO Simple_Account_View VALUES (1...	1 row(s) affected	0.016 sec
43	22:18:43	SELECT * FROM Account LIMIT 0. 1000	11 row(s) returned	0.000 sec / 0.000 sec
44	22:19:13	DELETE FROM Simple_Account_View WHERE ...	1 row(s) affected	0.000 sec
45	22:19:19	SELECT * FROM Account LIMIT 0. 1000	10 row(s) returned	0.000 sec / 0.000 sec
46	22:19:52	SHOW FULL TABLES WHERE TABLE_TYPE = ...	32 row(s) returned	0.015 sec / 0.000 sec
47	22:20:33	SHOW CREATE VIEW Customer_Account_View	1 row(s) returned	0.000 sec / 0.000 sec
48	22:21:01	DESCRIBE Customer_Account_View	7 row(s) returned	0.000 sec / 0.000 sec
49	22:21:29	DROP VIEW Customer_View	0 row(s) affected	0.016 sec
50	22:22:06	DROP VIEW Customer_View	Error Code: 1051. Unknown table '2520090181_H...	0.000 sec
51	22:22:19	DROP VIEW Customer_Basic_View, Savings_Ac...	0 row(s) affected	0.016 sec

Skill-ACID Properties

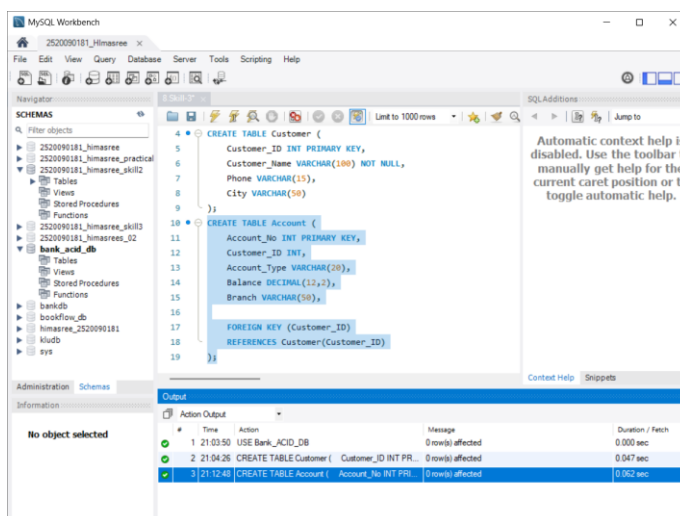
1. CREATE DATABASE



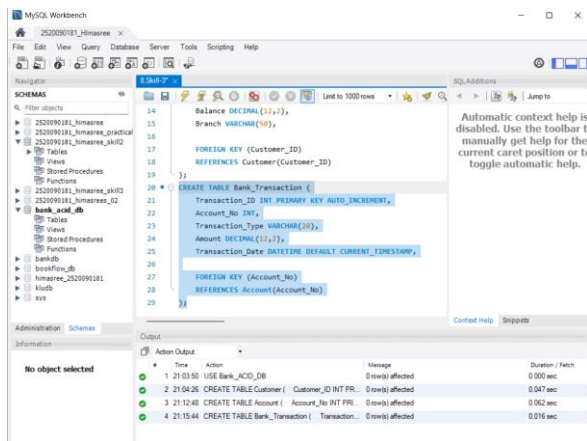
2. CREATE CUSTOMER TABLE



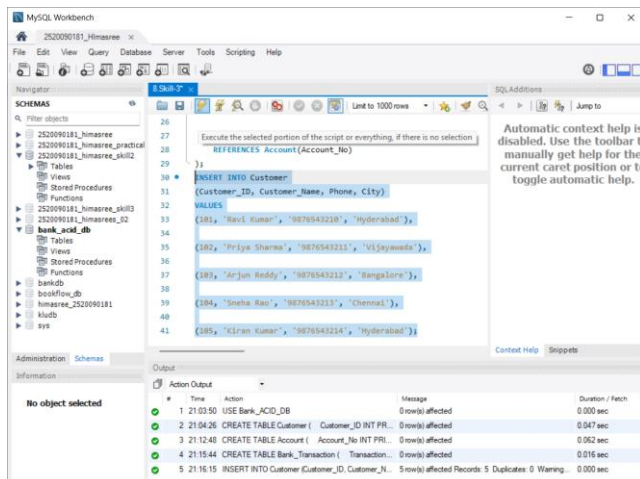
3. CREATE ACCOUNT TABLE



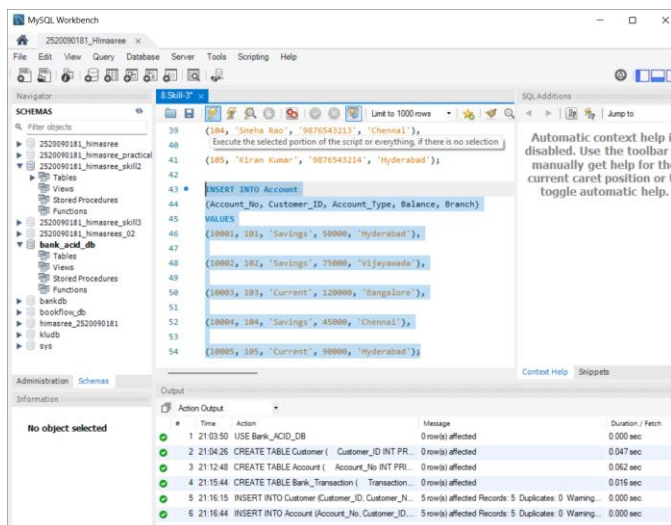
4. CREATE TRANSACTION TABLE



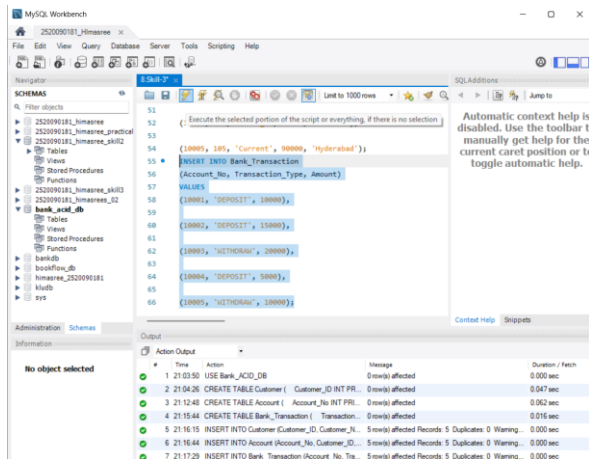
5. INSERT CUSTOMER DATA



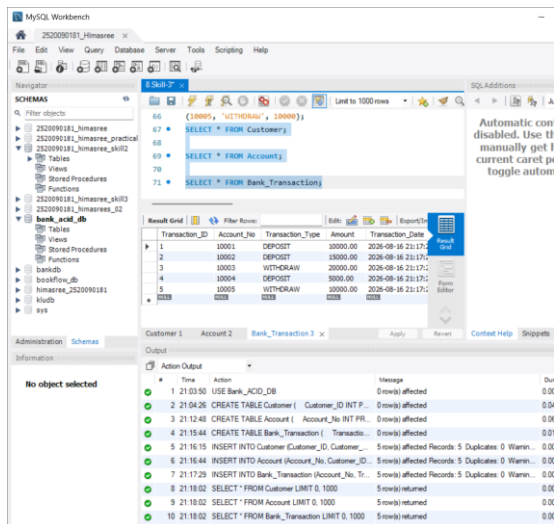
6. INSERT ACCOUNT DATA



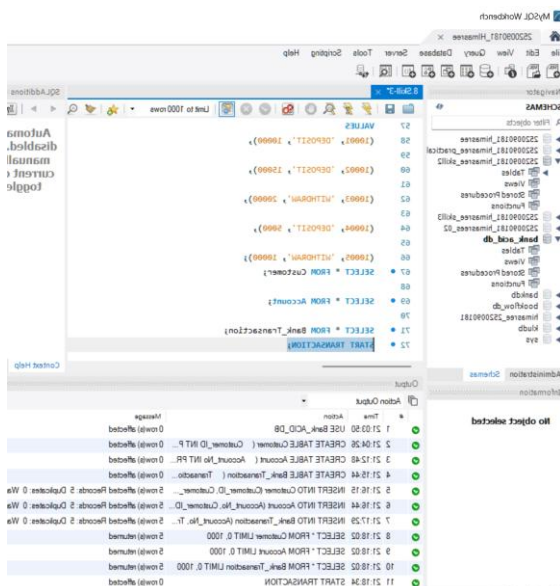
7. INSERT BANK TRANSACTION DATA



8. DISPLAY DATA



9. START A TRANSACTION



10. DEPOSIT MONEY

The screenshot shows the MySQL Workbench interface with the following SQL queries in the editor:

```
60 (10002, 'DEPOSIT', 15000);
61
62 (10003, 'WITHDRAW', 20000);
63
64 (10004, 'DEPOSIT', 5000);
65
66 (10005, 'WITHDRAW', 10000);
67
68 * FROM Customer;
69
70 * FROM Account;
71
72 * FROM Bank_Transaction;
73
74 START TRANSACTION;
75
76 UPDATE Account
77 SET Balance = Balance + 5000
78 WHERE Account_No = 10001;
```

The Output pane shows the execution results:

#	Time	Action	Message
2	21:04:26	CREATE TABLE Customer (Customer_ID INT ...	0 rows affected
3	21:12:48	CREATE TABLE Account (Account_No INT P...	0 rows affected
4	21:15:44	CREATE TABLE Bank_Transaction (Transac...	0 rows affected
5	21:16:15	INSERT INTO Customer (Customer_ID, Customer...	5 rows affected Records: 5 Duplicates: 0 Warnings: 0
6	21:16:44	INSERT INTO Account (Account_No, Customer...	5 rows affected Records: 5 Duplicates: 0 Warnings: 0
7	21:17:29	INSERT INTO Bank_Transaction (Account_No...	5 rows affected Records: 5 Duplicates: 0 Warnings: 0
8	21:18:02	SELECT * FROM Customer LIMIT 0, 1000	5 rows returned
9	21:18:02	SELECT * FROM Account LIMIT 0, 1000	5 rows returned
10	21:18:02	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 rows returned
11	21:18:34	START TRANSACTION	0 rows affected
12	21:23:06	UPDATE Account SET Balance = Balance + 500...	1 row(s) affected Rows matched: 1 Changed: 1

11. CHECK BALANCE

The screenshot shows the MySQL Workbench interface with the following SQL queries in the editor:

```
71 * SELECT * FROM Customer;
72
73 * SELECT * FROM Account;
74
75 * SELECT * FROM Bank_Transaction;
76
77 * SELECT * FROM Customer;
78
79 * SELECT * FROM Account;
80
81 * SELECT * FROM Bank_Transaction;
```

The Output pane shows the execution results:

#	Time	Action	Message
3	21:12:48	CREATE TABLE Account (Account_No INT P...	0 rows affected
4	21:15:44	CREATE TABLE Bank_Transaction (Transac...	0 rows affected
5	21:16:15	INSERT INTO Customer (Customer_ID, Customer...	5 rows affected Records: 5 Duplicates: 0 Warnings: 0
6	21:16:44	INSERT INTO Account (Account_No, Customer...	5 rows affected Records: 5 Duplicates: 0 Warnings: 0
7	21:17:29	INSERT INTO Bank_Transaction (Account_No...	5 rows affected Records: 5 Duplicates: 0 Warnings: 0
8	21:18:02	SELECT * FROM Customer LIMIT 0, 1000	5 rows returned
9	21:18:02	SELECT * FROM Account LIMIT 0, 1000	5 rows returned
10	21:18:02	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 rows returned
11	21:18:34	START TRANSACTION	0 rows affected
12	21:23:06	UPDATE Account SET Balance = Balance + 500...	1 row(s) affected Rows matched: 1 Changed: 1
13	21:23:35	SELECT * FROM Account WHERE Account_No...	1 row(s) returned

12. COMMIT TRANSACTION

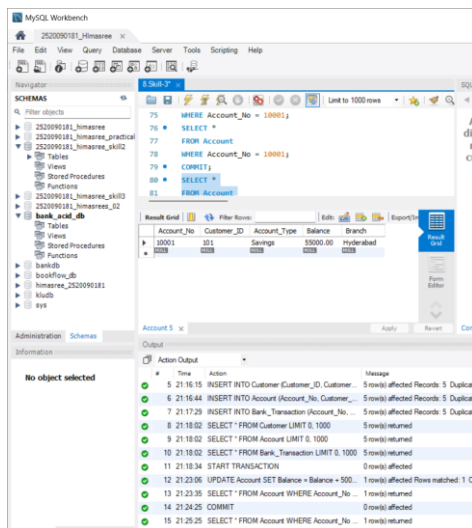
The screenshot shows the MySQL Workbench interface with the following SQL queries in the editor:

```
64 (10004, 'DEPOSIT', 5000);
65
66 (10005, 'WITHDRAW', 10000);
67
68 * FROM Customer;
69
70 * FROM Account;
71
72 * FROM Bank_Transaction;
73
74 START TRANSACTION;
75
76 UPDATE Account
77 SET Balance = Balance + 5000
78 WHERE Account_No = 10001;
79
80 COMMIT;
```

The Output pane shows the execution results:

#	Time	Action	Message	Duration / Fec
4	21:15:44	CREATE TABLE Bank_Transaction (Transac...	0 rows affected	0.015 sec
5	21:16:15	INSERT INTO Customer (Customer_ID, Customer...	5 rows affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
6	21:16:44	INSERT INTO Account (Account_No, Customer...	5 rows affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
7	21:17:29	INSERT INTO Bank_Transaction (Account_No...	5 rows affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
8	21:18:02	SELECT * FROM Customer LIMIT 0, 1000	5 rows returned	0.000 sec / 0.1
9	21:18:02	SELECT * FROM Account LIMIT 0, 1000	5 rows returned	0.000 sec / 0.1
10	21:18:02	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 rows returned	0.000 sec / 0.1
11	21:18:34	START TRANSACTION	0 rows affected	0.000 sec
12	21:23:06	UPDATE Account SET Balance = Balance + 500...	1 row(s) affected Rows matched: 1 Changed: 1	0.000 sec
13	21:23:35	SELECT * FROM Account WHERE Account_No...	1 row(s) returned	0.000 sec / 0.1
14	21:24:25	COMMIT	0 rows affected	0.015 sec

13. CHECK BALANCE AFTER COMMIT



The screenshot shows the MySQL Workbench interface. The query window contains the following SQL code:

```
75 WHERE Account_No = 10001;  
76 SELECT *  
77 FROM Account  
78 WHERE Account_No = 10001;  
79 COMMIT;  
80 SELECT *  
81 FROM Account
```

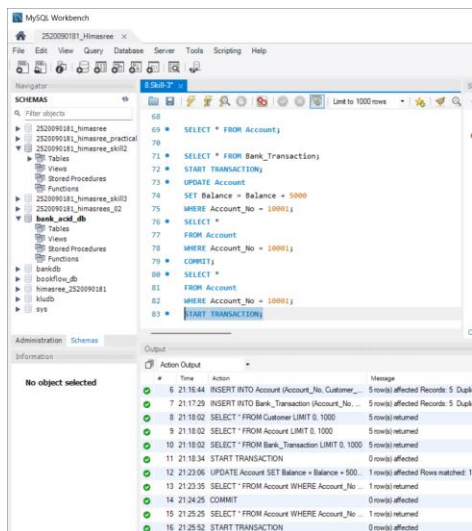
The output window shows the results of the query:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	5000.00	Hyderabad
10001	1001	1001	1001	1001

The Action Output window shows the following messages:

- 5 21:16:15 INSERT INTO Customer (Customer_ID, Customer_Name, Address, Contact_No) VALUES (1001, 'Kumar', '123 Main St', '9876543210') 5 row(s) affected Records: 5 Duplicate
- 6 21:16:44 INSERT INTO Account (Account_No, Customer_No, Balance, Branch) VALUES (10001, 1001, 5000.00, 'Hyderabad') 5 row(s) affected Records: 5 Duplicate
- 7 21:17:29 INSERT INTO Bank_Transaction (Account_No, Customer_No, Transaction_No, Amount, Transaction_Type) VALUES (10001, 1001, 1001, 1000, 'Deposit') 5 row(s) affected Records: 5 Duplicate
- 8 21:18:02 SELECT * FROM Customer LIMIT 0, 1000 5 row(s) returned
- 9 21:18:02 SELECT * FROM Account LIMIT 0, 1000 5 row(s) returned
- 10 21:18:02 SELECT * FROM Bank_Transaction LIMIT 0, 1000 5 row(s) returned
- 11 21:18:34 START TRANSACTION 0 row(s) affected
- 12 21:23:06 UPDATE Account SET Balance = Balance + 500. 1 row(s) affected Rows matched: 1 Ch
- 13 21:23:35 SELECT * FROM Account WHERE Account_No = 10001 1 row(s) returned
- 14 21:24:25 COMMIT 0 row(s) affected
- 15 21:25:25 SELECT * FROM Account WHERE Account_No = 10001 1 row(s) returned

14. START TRANSACTION



The screenshot shows the MySQL Workbench interface. The query window contains the following SQL code:

```
68 SELECT * FROM Account;  
69  
70  
71 SELECT * FROM Bank_Transaction;  
72  
73 START TRANSACTION;  
74  
75 UPDATE Account  
76 SET Balance = Balance + 5000  
77 WHERE Account_No = 10001;  
78  
79 SELECT *  
80 FROM Account  
81 WHERE Account_No = 10001;  
82  
83 COMMIT;  
84  
85 SELECT *  
86 FROM Account  
87 WHERE Account_No = 10001;  
88  
89 START TRANSACTION;
```

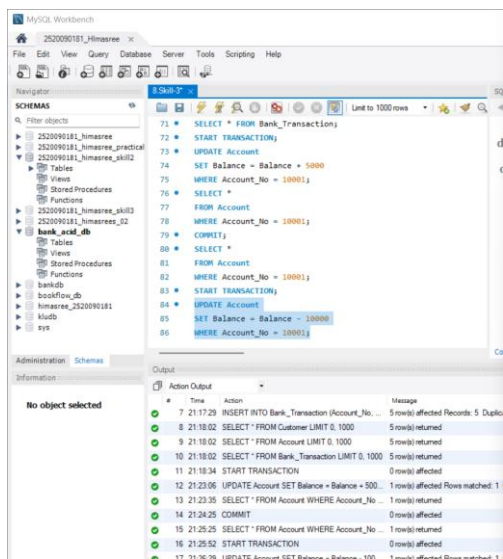
The output window shows the results of the query:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	5000.00	Hyderabad
10001	1001	1001	1001	1001

The Action Output window shows the following messages:

- 6 21:16:44 INSERT INTO Account (Account_No, Customer_No, Balance, Branch) VALUES (10001, 1001, 5000.00, 'Hyderabad') 5 row(s) affected Records: 5 Dupl
- 7 21:17:29 INSERT INTO Bank_Transaction (Account_No, Customer_No, Transaction_No, Amount, Transaction_Type) VALUES (10001, 1001, 1001, 1000, 'Deposit') 5 row(s) affected Records: 5 Dupl
- 8 21:18:02 SELECT * FROM Customer LIMIT 0, 1000 5 row(s) returned
- 9 21:18:02 SELECT * FROM Account LIMIT 0, 1000 5 row(s) returned
- 10 21:18:02 SELECT * FROM Bank_Transaction LIMIT 0, 1000 5 row(s) returned
- 11 21:18:34 START TRANSACTION 0 row(s) affected
- 12 21:23:06 UPDATE Account SET Balance = Balance + 500. 1 row(s) affected Rows matched: 1
- 13 21:23:35 SELECT * FROM Account WHERE Account_No = 10001 1 row(s) returned
- 14 21:24:25 COMMIT 0 row(s) affected
- 15 21:25:25 SELECT * FROM Account WHERE Account_No = 10001 1 row(s) returned
- 16 21:25:52 START TRANSACTION 0 row(s) affected

15. WITHDRAW MONEY



The screenshot shows the MySQL Workbench interface. The query window contains the following SQL code:

```
71 SELECT * FROM Bank_Transaction;  
72  
73 START TRANSACTION;  
74  
75 UPDATE Account  
76 SET Balance = Balance + 5000  
77 WHERE Account_No = 10001;  
78  
79 SELECT *  
80 FROM Account  
81 WHERE Account_No = 10001;  
82  
83 COMMIT;  
84  
85 SELECT *  
86 FROM Account  
87 WHERE Account_No = 10001;  
88  
89 START TRANSACTION;  
90  
91 UPDATE Account  
92 SET Balance = Balance - 10000  
93 WHERE Account_No = 10001;
```

The output window shows the results of the query:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	5000.00	Hyderabad
10001	1001	1001	1001	1001

The Action Output window shows the following messages:

- 7 21:17:29 INSERT INTO Bank_Transaction (Account_No, Customer_No, Transaction_No, Amount, Transaction_Type) VALUES (10001, 1001, 1001, 1000, 'Deposit') 5 row(s) affected Records: 5 Dupl
- 8 21:18:02 SELECT * FROM Customer LIMIT 0, 1000 5 row(s) returned
- 9 21:18:02 SELECT * FROM Account LIMIT 0, 1000 5 row(s) returned
- 10 21:18:02 SELECT * FROM Bank_Transaction LIMIT 0, 1000 5 row(s) returned
- 11 21:18:34 START TRANSACTION 0 row(s) affected
- 12 21:23:06 UPDATE Account SET Balance = Balance + 500. 1 row(s) affected Rows matched: 1
- 13 21:23:35 SELECT * FROM Account WHERE Account_No = 10001 1 row(s) returned
- 14 21:24:25 COMMIT 0 row(s) affected
- 15 21:25:25 SELECT * FROM Account WHERE Account_No = 10001 1 row(s) returned
- 16 21:25:52 START TRANSACTION 0 row(s) affected
- 17 21:26:25 UPDATE Account SET Balance = Balance - 100. 1 row(s) affected Rows matched: 1

16. CHECK BALANCE

The screenshot shows a MySQL Workbench window with a SQL script in the editor. The script includes a WHERE clause, a START TRANSACTION statement, an UPDATE statement, a SET statement, a WHERE clause, and a SELECT statement. The output pane displays a table with columns: Account_No, Customer_ID, Account_Type, Balance, and Branch. The table contains two rows of data.

```
82 WHERE Account_No = 10001;
83 START TRANSACTION;
84 UPDATE Account
85 SET Balance = Balance - 10000
86 WHERE Account_No = 10001;
87 SELECT *
88 FROM Account;
```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	45000.00	Hyderabad
10008	009	0008	0008	

17. ROLLBACK

The screenshot shows a MySQL Workbench window with a SQL script in the editor. The script includes a WHERE clause, a SELECT statement, a FROM clause, a WHERE clause, a COMMIT statement, a FROM clause, a WHERE clause, a START TRANSACTION statement, an UPDATE statement, a SET statement, a WHERE clause, a SELECT statement, a FROM clause, a WHERE clause, and a ROLLBACK statement. The output pane displays a table with columns: #, Time, Action, and Message. The table contains 20 rows of data, including the ROLLBACK statement.

```
75 WHERE Account_No = 10001;
76 SELECT *
77 FROM Account
78 WHERE Account_No = 10001;
79 COMMIT;
80 SELECT *
81 FROM Account
82 WHERE Account_No = 10001;
83 START TRANSACTION;
84 UPDATE Account
85 SET Balance = Balance - 10000
86 WHERE Account_No = 10001;
87 SELECT *
88 FROM Account
89 WHERE Account_No = 10001;
90 ROLLBACK;
```

#	Time	Action	Message
8	21:18:02	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
9	21:18:02	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
10	21:18:02	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
11	21:18:04	START TRANSACTION	0 row(s) affected
12	21:23:06	UPDATE Account SET Balance = Balance - 500.	1 row(s) affected Rows matched: 1 Changed: 1
13	21:23:35	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned
14	21:24:25	COMMIT	0 row(s) affected
15	21:25:25	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned
16	21:25:52	START TRANSACTION	0 row(s) affected
17	21:26:29	UPDATE Account SET Balance = Balance - 100.	1 row(s) affected Rows matched: 1 Changed: 1
18	21:27:45	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned
19	21:28:08	WHERE Account_No = 10001	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'WHERE Account_No = 10001' at line 1
20	21:28:12	ROLLBACK	0 row(s) affected

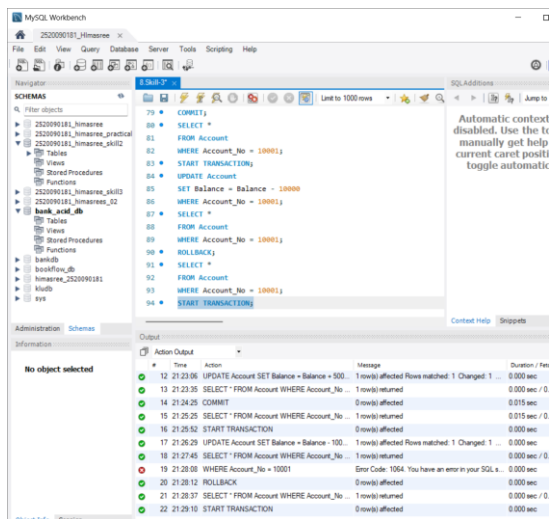
18. CHECK BALANCE AFTER ROLLBACK

The screenshot shows a MySQL Workbench window with a SQL script in the editor. The script includes a WHERE clause, a SELECT statement, a FROM clause, a WHERE clause, a ROLLBACK statement, a SELECT statement, and a FROM clause. The output pane displays a table with columns: #, Time, Action, Message, and Duration / Row(s). The table contains 21 rows of data, including the ROLLBACK statement.

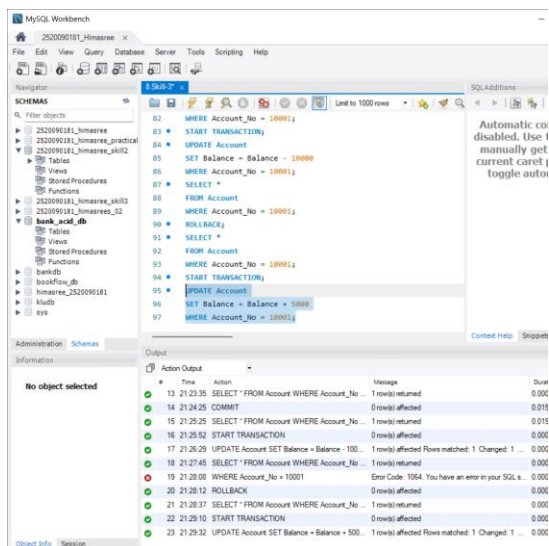
```
86 WHERE Account_No = 10001;
87 * Execute the selected portion of the script or everything, if there is no selection
88 FROM Account;
89 WHERE Account_No = 10001;
90 ROLLBACK;
91 SELECT *
92 FROM Account;
```

#	Time	Action	Message	Duration / Row(s)
11	21:18:04	START TRANSACTION	0 row(s) affected	0.000 sec
12	21:23:06	UPDATE Account SET Balance = Balance - 500.	1 row(s) affected Rows matched: 1 Changed: 1	0.000 sec
13	21:23:35	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned	0.000 sec / 0.000 sec
14	21:24:25	COMMIT	0 row(s) affected	0.015 sec
15	21:25:25	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned	0.015 sec / 0.000 sec
16	21:25:52	START TRANSACTION	0 row(s) affected	0.000 sec
17	21:26:29	UPDATE Account SET Balance = Balance - 100.	1 row(s) affected Rows matched: 1 Changed: 1	0.000 sec
18	21:27:45	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned	0.000 sec / 0.000 sec
19	21:28:08	WHERE Account_No = 10001	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'WHERE Account_No = 10001' at line 1	0.000 sec
20	21:28:12	ROLLBACK	0 row(s) affected	0.000 sec
21	21:28:37	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned	0.000 sec / 0.000 sec

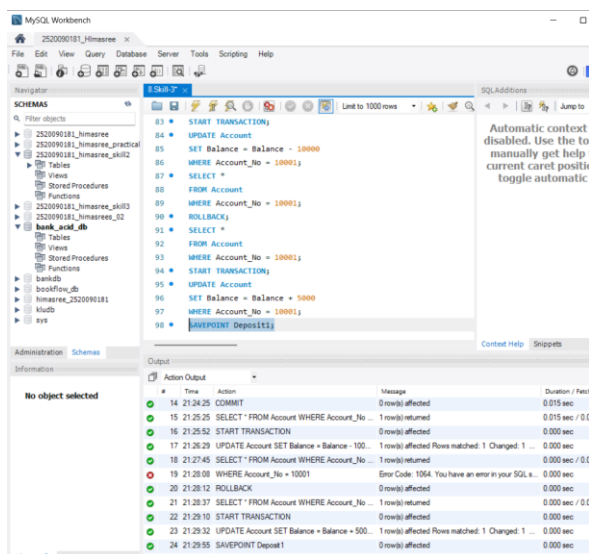
19. START TRANSACTION



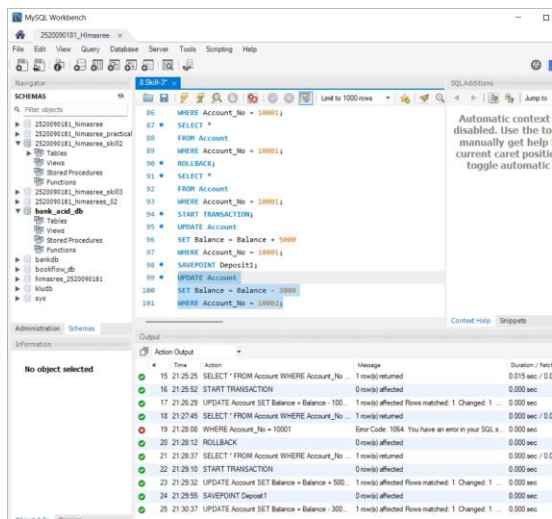
20. FIRST UPDATE



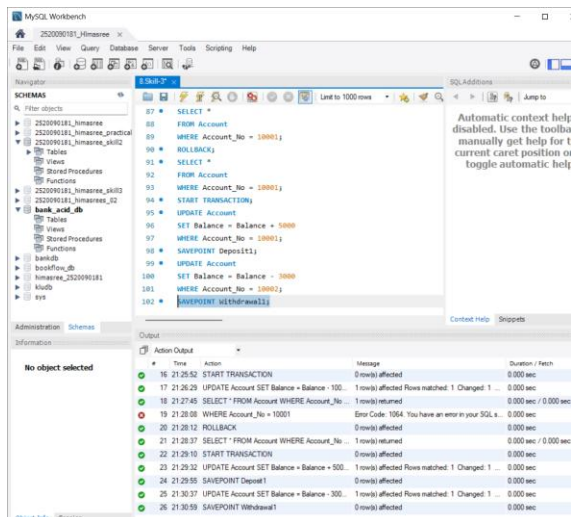
21. CREATE SAVEPOINT



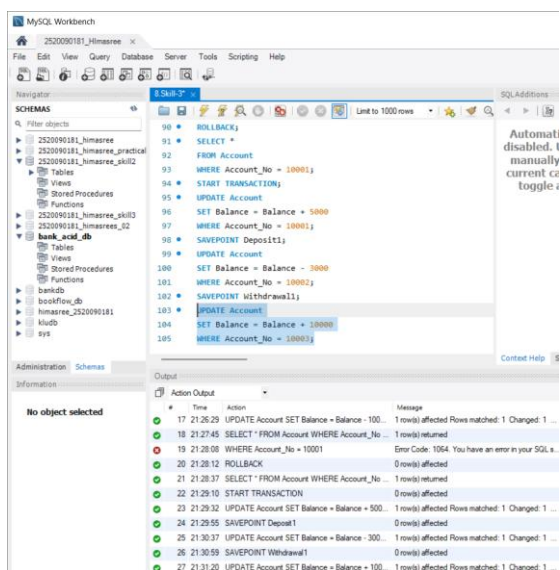
22. SECOND UPDATE



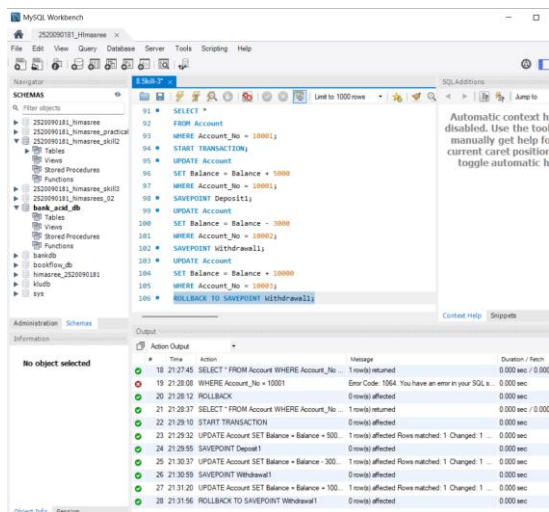
23. CREATE SECOND SAVEPOINT



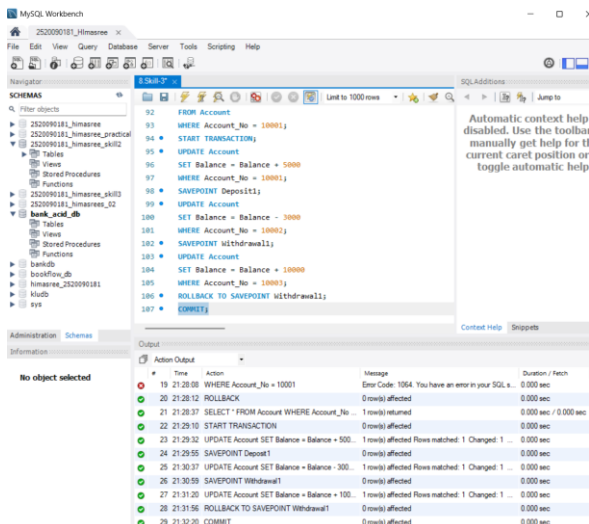
24. THIRD UPDATE



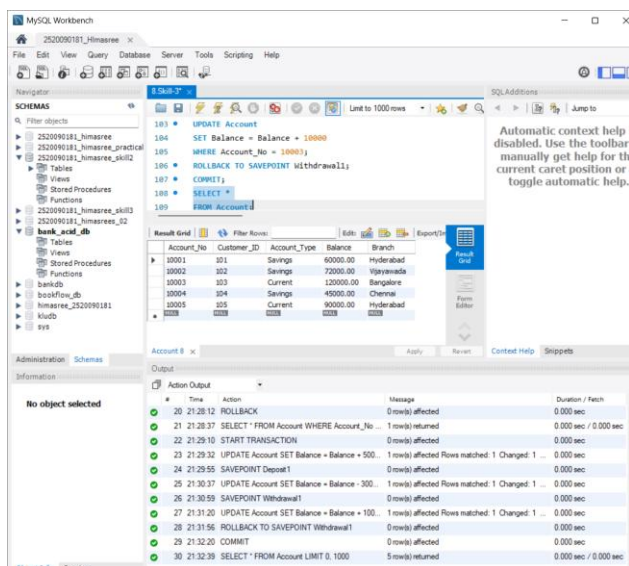
25. ROLLBACK TO SAVEPOINT



26. COMMIT



27. CHECK ACCOUNTS



28. TRANSFER MONEY

MySQL Workbench interface showing a SQL script for a money transfer transaction. The script is as follows:

```
96 SET Balance = Balance + 5000
97 WHERE Account_No = 10001;
98 SAVEPOINT Deposit1;
99 UPDATE Account
100 SET Balance = Balance - 3000
101 WHERE Account_No = 10001;
102 SAVEPOINT Withdrawal1;
103 UPDATE Account
104 SET Balance = Balance + 10000
105 WHERE Account_No = 10001;
106 ROLLBACK TO SAVEPOINT Withdrawal1;
107 COMMIT;
108 SELECT *
109 FROM Account;
110 START TRANSACTION;
```

The output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
21	21:29:37	SELECT * FROM Account WHERE Account_No = 10001	5 row(s) returned	0.000 sec / 0.000 sec
22	21:29:10	START TRANSACTION	0 row(s) affected	0.000 sec
23	21:29:32	UPDATE Account SET Balance = Balance + 5000	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
24	21:29:55	SAVEPOINT Deposit1	0 row(s) affected	0.000 sec
25	21:30:37	UPDATE Account SET Balance = Balance - 3000	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
26	21:30:59	SAVEPOINT Withdrawal1	0 row(s) affected	0.000 sec
27	21:31:20	UPDATE Account SET Balance = Balance + 10000	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
28	21:31:56	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected	0.000 sec
29	21:32:20	COMMIT	0 row(s) affected	0.000 sec
30	21:32:39	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
31	21:33:04	START TRANSACTION	0 row(s) affected	0.000 sec

29. DEDUCT FROM SENDER

MySQL Workbench interface showing a SQL script for deducting money from the sender. The script is as follows:

```
98 SAVEPOINT Deposit1;
99 UPDATE Account
100 SET Balance = Balance - 3000
101 WHERE Account_No = 10001;
102 SAVEPOINT Withdrawal1;
103 UPDATE Account
104 SET Balance = Balance + 10000
105 WHERE Account_No = 10001;
106 ROLLBACK TO SAVEPOINT Withdrawal1;
107 COMMIT;
108 SELECT *
109 FROM Account;
110 START TRANSACTION;
111 UPDATE Account
112 SET Balance = Balance - 10000
113 WHERE Account_No = 10001;
```

The output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
22	21:29:10	START TRANSACTION	0 row(s) affected	0.000 sec
23	21:29:32	UPDATE Account SET Balance = Balance + 5000	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
24	21:29:55	SAVEPOINT Deposit1	0 row(s) affected	0.000 sec
25	21:30:37	UPDATE Account SET Balance = Balance - 3000	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
26	21:30:59	SAVEPOINT Withdrawal1	0 row(s) affected	0.000 sec
27	21:31:20	UPDATE Account SET Balance = Balance + 10000	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
28	21:31:56	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected	0.000 sec
29	21:32:20	COMMIT	0 row(s) affected	0.000 sec
30	21:32:39	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
31	21:33:04	START TRANSACTION	0 row(s) affected	0.000 sec
32	21:33:25	UPDATE Account SET Balance = Balance - 10000	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec

30. ADD TO RECEIVER

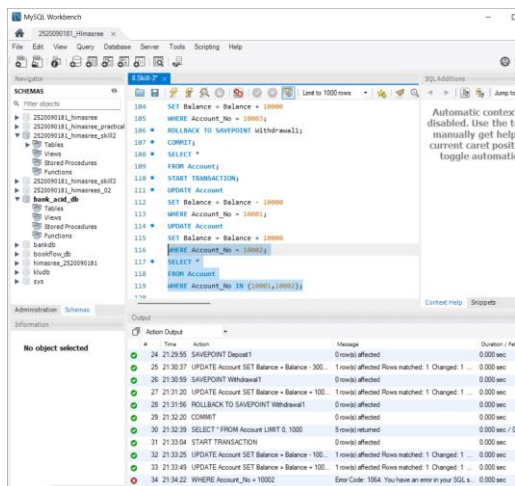
MySQL Workbench interface showing a SQL script for adding money to the receiver. The script is as follows:

```
102 SAVEPOINT Withdrawal1;
103 UPDATE Account
104 SET Balance = Balance - 10000
105 WHERE Account_No = 10001;
106 ROLLBACK TO SAVEPOINT Withdrawal1;
107 COMMIT;
108 SELECT *
109 FROM Account;
110 START TRANSACTION;
111 UPDATE Account
112 SET Balance = Balance + 10000
113 WHERE Account_No = 10001;
114 UPDATE Account
115 SET Balance = Balance + 10000
116 WHERE Account_No = 10001;
```

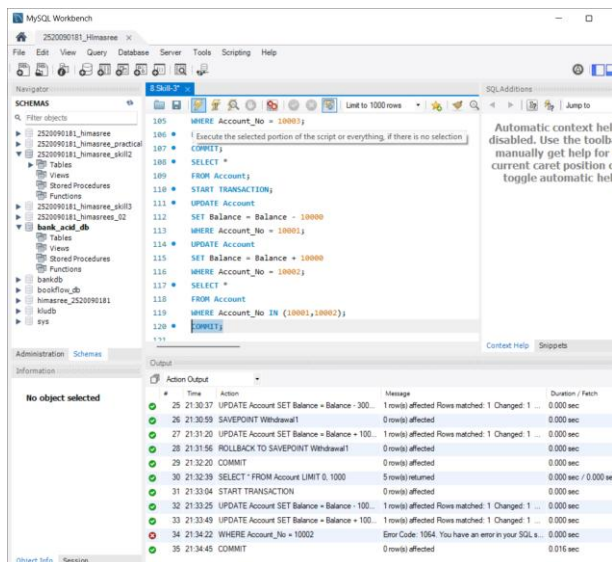
The output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
23	21:29:32	UPDATE Account SET Balance = Balance + 5000	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
24	21:29:55	SAVEPOINT Deposit1	0 row(s) affected	0.000 sec
25	21:30:37	UPDATE Account SET Balance = Balance - 3000	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
26	21:30:59	SAVEPOINT Withdrawal1	0 row(s) affected	0.000 sec
27	21:31:20	UPDATE Account SET Balance = Balance + 10000	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
28	21:31:56	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected	0.000 sec
29	21:32:20	COMMIT	0 row(s) affected	0.000 sec
30	21:32:39	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
31	21:33:04	START TRANSACTION	0 row(s) affected	0.000 sec
32	21:33:25	UPDATE Account SET Balance = Balance - 10000	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
33	21:33:43	UPDATE Account SET Balance = Balance + 10000	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec

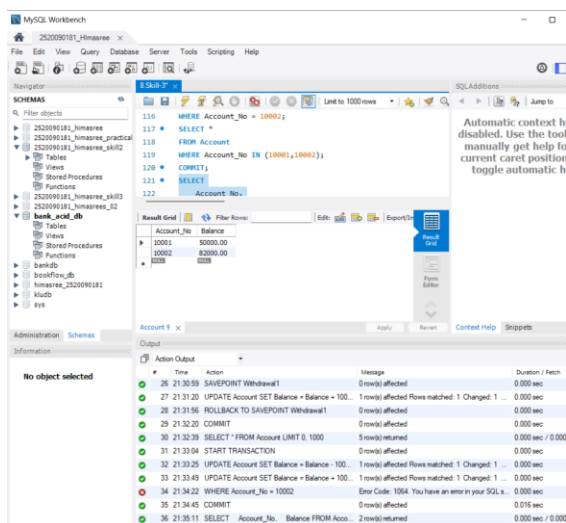
31. CHECK BOTH ACCOUNTS



32. COMMIT TRANSFER



33. FINAL BALANCES



34. START TRANSACTION

MySQL Workbench interface showing a SQL query in the Editor. The query is:

```
111 UPDATE Account
112 SET Balance = Balance - 10000
113 WHERE Account_No = 10002;
114 UPDATE Account
115 SET Balance = Balance + 10000
116 WHERE Account_No = 10002;
117 SELECT *
118 FROM Account
119 WHERE Account_No IN (10001,10002);
120 COMMIT;
121 SELECT
122     Account_No,
123     Balance
124 FROM Account
125 WHERE Account_No IN (10001,10002);
126 START TRANSACTION;
```

The Output window shows the execution results:

#	Time	Action	Message	Duration / Fech
27	21:31:20	UPDATE Account SET Balance = Balance + 100...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.00
28	21:31:56	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected	0.00
29	21:32:20	COMMIT	0 row(s) affected	0.00
30	21:32:39	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.00
31	21:33:04	START TRANSACTION	0 row(s) affected	0.00
32	21:33:25	UPDATE Account SET Balance = Balance - 100...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.00
33	21:33:49	UPDATE Account SET Balance = Balance + 100...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.00
34	21:34:22	WHERE Account_No = 10002	Error Code: 1064: You have an error in your SQL s...	0.00
35	21:34:45	COMMIT	0 row(s) affected	0.01
36	21:35:11	SELECT Account_No, Balance FROM Acco...	2 row(s) returned	0.00
37	21:35:33	START TRANSACTION	0 row(s) affected	0.00

Automatic context help is disabled. Use manually get current caret position to toggle auto.

35. DEDUCT MONEY

MySQL Workbench interface showing a SQL query in the Editor. The query is:

```
114 UPDATE Account
115 SET Balance = Balance + 10000
116 WHERE Account_No = 10002;
117 SELECT *
118 FROM Account
119 WHERE Account_No IN (10001,10002);
120 COMMIT;
121 SELECT
122     Account_No,
123     Balance
124 FROM Account
125 WHERE Account_No IN (10001,10002);
126 START TRANSACTION;
127 UPDATE Account
128 SET Balance = Balance - 20000
129 WHERE Account_No = 10001;
```

The Output window shows the execution results:

#	Time	Action	Message	Duration / Fech
28	21:31:56	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected	0.000 sec
29	21:32:20	COMMIT	0 row(s) affected	0.000 sec
30	21:32:39	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000
31	21:33:04	START TRANSACTION	0 row(s) affected	0.000 sec
32	21:33:25	UPDATE Account SET Balance = Balance - 100...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
33	21:33:49	UPDATE Account SET Balance = Balance + 100...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
34	21:34:22	WHERE Account_No = 10002	Error Code: 1064: You have an error in your SQL s...	0.000 sec
35	21:34:45	COMMIT	0 row(s) affected	0.016 sec
36	21:35:11	SELECT Account_No, Balance FROM Acco...	2 row(s) returned	0.000 sec / 0.000
37	21:35:33	START TRANSACTION	0 row(s) affected	0.000 sec
38	21:35:55	UPDATE Account SET Balance = Balance - 200...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec

Automatic context help is disabled. Use the toolbar manually get help for current caret position to toggle automatic help.

36. ADD MONEY

MySQL Workbench interface showing a SQL query in the Editor. The query is:

```
117 SELECT *
118 FROM Account
119 WHERE Account_No IN (10001,10002);
120 COMMIT;
121 SELECT
122     Account_No,
123     Balance
124 FROM Account
125 WHERE Account_No IN (10001,10002);
126 START TRANSACTION;
127 UPDATE Account
128 SET Balance = Balance - 20000
129 WHERE Account_No = 10001;
130 UPDATE Account
131 SET Balance = Balance + 20000
132 WHERE Account_No = 10002;
```

The Output window shows the execution results:

#	Time	Action	Message	Duration / Fech
29	21:32:20	COMMIT	0 row(s) affected	0.000 sec
30	21:32:39	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
31	21:33:04	START TRANSACTION	0 row(s) affected	0.000 sec
32	21:33:25	UPDATE Account SET Balance = Balance - 100...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
33	21:33:49	UPDATE Account SET Balance = Balance + 100...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
34	21:34:22	WHERE Account_No = 10002	Error Code: 1064: You have an error in your SQL s...	0.000 sec
35	21:34:45	COMMIT	0 row(s) affected	0.016 sec
36	21:35:11	SELECT Account_No, Balance FROM Acco...	2 row(s) returned	0.000 sec / 0.000 sec
37	21:35:33	START TRANSACTION	0 row(s) affected	0.000 sec
38	21:35:55	UPDATE Account SET Balance = Balance - 200...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
39	21:36:16	UPDATE Account SET Balance = Balance + 200...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec

Automatic context help is disabled. Use the toolbar manually get help for current caret position to toggle automatic help.

37. CANCEL TRANSACTION

MySQL Workbench interface showing a transaction being rolled back. The SQL editor contains the following code:

```
118 FROM Account
119 WHERE Account_No IN (10001,10002);
120 * COMMIT;
121 * SELECT
122     Account_No,
123     Balance
124 FROM Account
125 WHERE Account_No IN (10001,10002);
126 * START TRANSACTION;
127 * UPDATE Account
128 SET Balance = Balance - 20000
129 WHERE Account_No = 10001;
130 * UPDATE Account
131 SET Balance = Balance + 20000
132 WHERE Account_No = 10002;
133 * ROLLBACK;
```

The Output pane shows the execution results:

#	Time	Action	Message	Duration / Fetch
30	21:32:39	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000
31	21:33:04	START TRANSACTION	0 row(s) affected	0.000 sec
32	21:33:25	UPDATE Account SET Balance = Balance - 100;	1 row(s) affected Rows matched: 1 Changed: 1	0.000 sec
33	21:33:45	UPDATE Account SET Balance = Balance + 100;	1 row(s) affected Rows matched: 1 Changed: 1	0.000 sec
34	21:34:02	WHERE Account_No = 10002	Error Code: 1064: You have an error in your SQL s...	0.000 sec
35	21:34:45	COMMIT	0 row(s) affected	0.016 sec
36	21:35:11	SELECT Account_No, Balance FROM Acco...	2 row(s) returned	0.000 sec / 0.000
37	21:35:33	START TRANSACTION	0 row(s) affected	0.000 sec
38	21:35:55	UPDATE Account SET Balance = Balance - 200;	1 row(s) affected Rows matched: 1 Changed: 1	0.000 sec
39	21:36:16	UPDATE Account SET Balance = Balance + 200;	1 row(s) affected Rows matched: 1 Changed: 1	0.000 sec
40	21:36:55	ROLLBACK	0 row(s) affected	0.000 sec

38. CHECK BALANCES

MySQL Workbench interface showing a query result grid. The SQL editor contains the following code:

```
129 WHERE Account_No = 10001;
130 * UPDATE Account
131 SET Balance = Balance + 20000
132 WHERE Account_No = 10002;
133 * ROLLBACK;
134 * SELECT *
135 FROM Account;
```

The Result Grid shows the following data:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	82000.00	Vijayawada

The Output pane shows the execution results:

#	Time	Action	Message	Duration / Fetch
31	21:33:04	START TRANSACTION	0 row(s) affected	0.000 sec
32	21:33:25	UPDATE Account SET Balance = Balance - 100;	1 row(s) affected Rows matched: 1 Changed: 1	0.000 sec
33	21:33:45	UPDATE Account SET Balance = Balance + 100;	1 row(s) affected Rows matched: 1 Changed: 1	0.000 sec
34	21:34:22	WHERE Account_No = 10002	Error Code: 1064: You have an error in your SQL s...	0.000 sec
35	21:34:45	COMMIT	0 row(s) affected	0.016 sec
36	21:35:11	SELECT Account_No, Balance FROM Acco...	2 row(s) returned	0.000 sec / 0.000 sec
37	21:35:33	START TRANSACTION	0 row(s) affected	0.000 sec
38	21:35:55	UPDATE Account SET Balance = Balance - 200;	1 row(s) affected Rows matched: 1 Changed: 1	0.000 sec
39	21:36:16	UPDATE Account SET Balance = Balance + 200;	1 row(s) affected Rows matched: 1 Changed: 1	0.000 sec
40	21:36:55	ROLLBACK	0 row(s) affected	0.000 sec
41	21:37:14	SELECT * FROM Account WHERE Account_No...	2 row(s) returned	0.000 sec / 0.000 sec

39. ATOMICITY

MySQL Workbench interface showing a transaction being committed. The SQL editor contains the following code:

```
130 * UPDATE Account
131 SET Balance = Balance + 20000
132 WHERE Account_No = 10002;
133 * ROLLBACK;
134 * SELECT *
135 FROM Account;
136 WHERE Account_No IN (10001,10002);
```

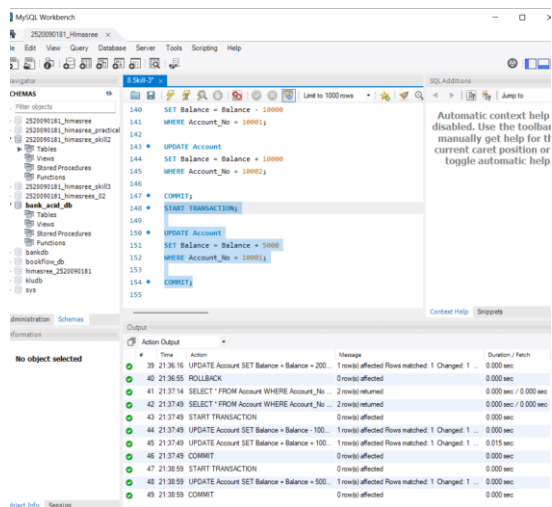
The Result Grid shows the following data:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	82000.00	Vijayawada

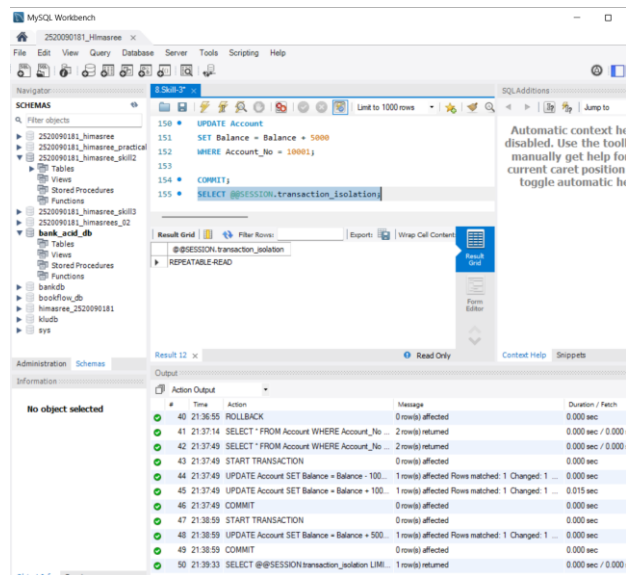
The Output pane shows the execution results:

#	Time	Action	Message	Duration / Fetch
36	21:35:11	SELECT Account_No, Balance FROM Acco...	2 row(s) returned	0.000 sec / 0.000
37	21:35:33	START TRANSACTION	0 row(s) affected	0.000 sec
38	21:35:55	UPDATE Account SET Balance = Balance - 200;	1 row(s) affected Rows matched: 1 Changed: 1	0.000 sec
39	21:36:16	UPDATE Account SET Balance = Balance + 200;	1 row(s) affected Rows matched: 1 Changed: 1	0.000 sec
40	21:36:55	ROLLBACK	0 row(s) affected	0.000 sec
41	21:37:14	SELECT * FROM Account WHERE Account_No...	2 row(s) returned	0.000 sec / 0.000
42	21:37:49	SELECT * FROM Account WHERE Account_No...	2 row(s) returned	0.000 sec / 0.000
43	21:37:49	START TRANSACTION	0 row(s) affected	0.000 sec
44	21:37:49	UPDATE Account SET Balance = Balance - 100;	1 row(s) affected Rows matched: 1 Changed: 1	0.000 sec
45	21:37:49	UPDATE Account SET Balance = Balance - 100;	1 row(s) affected Rows matched: 1 Changed: 1	0.016 sec
46	21:37:49	COMMIT	0 row(s) affected	0.000 sec

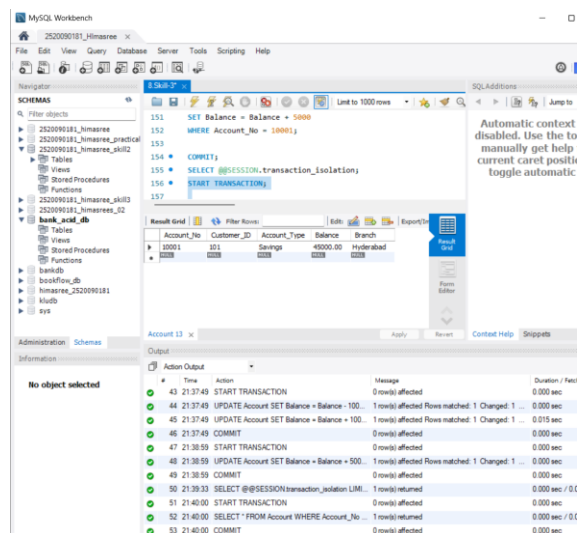
42. DURABILITY



43. CHECK CURRENT ISOLATION LEVEL



44. READ UNCOMMITTED



45. READ COMMITTED

MySQL Workbench interface showing the Read Committed isolation level. The SQL editor contains the following queries:

```
158 FROM Account
161 WHERE Account_No = 10001;
162
164 *
165 COMMIT;
166 START TRANSACTION;
```

The Result Grid shows one row:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	45000.00	Hyderabad

The Action Output pane shows the following sequence of actions:

Time	Action	Message	Duration
46 21:37:43	COMMIT	0 rows affected	0.000 sec
47 21:38:59	START TRANSACTION	0 rows affected	0.000 sec
48 21:38:59	UPDATE Account SET Balance = Balance + 500.	1 row(s) affected Rows matched: 1 Changed: 1	0.000 sec
49 21:38:59	COMMIT	0 rows affected	0.000 sec
50 21:39:33	SELECT @@SESSION.transaction_isolation (LM.	1 row(s) returned	0.000 sec
51 21:40:00	START TRANSACTION	0 rows affected	0.000 sec
52 21:40:00	SELECT * FROM Account WHERE Account_No...	1 row(s) returned	0.000 sec
53 21:40:00	COMMIT	0 rows affected	0.000 sec
54 21:40:40	START TRANSACTION	0 rows affected	0.015 sec
55 21:40:40	SELECT * FROM Account WHERE Account_No...	1 row(s) returned	0.000 sec
56 21:40:40	COMMIT	0 rows affected	0.000 sec

Automatic context disabled. Use the manually get the current caret position to toggle automatic.

46. REPEATABLE READ

MySQL Workbench interface showing the Repeatable Read isolation level. The SQL editor contains the following queries:

```
169 FROM Account
170 WHERE Account_No = 10001;
171
173 *
174 COMMIT;
175 START TRANSACTION;
```

The Result Grid shows one row:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	45000.00	Hyderabad

The Action Output pane shows the following sequence of actions:

Time	Action	Message	Duration
49 21:38:59	COMMIT	0 rows affected	0.000 sec
50 21:39:33	SELECT @@SESSION.transaction_isolation (LM.	1 row(s) returned	0.000 sec
51 21:40:00	START TRANSACTION	0 rows affected	0.000 sec
52 21:40:00	SELECT * FROM Account WHERE Account_No...	1 row(s) returned	0.000 sec
53 21:40:00	COMMIT	0 rows affected	0.015 sec
54 21:40:40	START TRANSACTION	1 row(s) returned	0.000 sec
55 21:40:40	COMMIT	0 rows affected	0.000 sec
56 21:40:40	COMMIT	0 rows affected	0.000 sec
57 21:41:04	START TRANSACTION	0 rows affected	0.000 sec
58 21:41:04	SELECT * FROM Account WHERE Account_No...	1 row(s) returned	0.000 sec
59 21:41:05	COMMIT	0 rows affected	0.000 sec

Automatic context disabled. Use the manually get the current caret position to toggle automatic.

47. SERIALIZABLE

MySQL Workbench interface showing the Serializable isolation level. The SQL editor contains the following queries:

```
176 FROM Account
179 WHERE Account_No = 10001;
180
182 *
183 COMMIT;
184 START TRANSACTION;
```

The Result Grid shows one row:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	45000.00	Hyderabad

The Action Output pane shows the following sequence of actions:

Time	Action	Message	Duration
52 21:40:00	SELECT * FROM Account WHERE Account_No...	1 row(s) returned	0.0
53 21:40:00	COMMIT	0 rows affected	0.0
54 21:40:40	START TRANSACTION	0 rows affected	0.0
55 21:40:40	SELECT * FROM Account WHERE Account_No...	1 row(s) returned	0.0
56 21:40:40	COMMIT	0 rows affected	0.0
57 21:41:04	START TRANSACTION	0 rows affected	0.0
58 21:41:04	SELECT * FROM Account WHERE Account_No...	1 row(s) returned	0.0
59 21:41:05	COMMIT	0 rows affected	0.0
60 21:41:43	START TRANSACTION	0 rows affected	0.0
61 21:41:43	SELECT * FROM Account WHERE Account_No...	1 row(s) returned	0.0
62 21:41:43	COMMIT	0 rows affected	0.0

Automatic context disabled. Use the manually get the current caret position to toggle automatic.

48. DEPOSIT TRANSACTION

MySQL Workbench

2520090181_himasee

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

SCHEMAS

- 2520090181_himasee
- 2520090181_himasee_practical
- 2520090181_himasee_skill2
- Tables
- Views
- Stored Procedures
- Functions
- 2520090181_himasee_skill3
- 2520090181_himasee_skill3
- bank_acad_db
- Tables
- Views
- Stored Procedures
- Functions
- bank_db
- bookflow_db
- himasee_2520090181
- kludb
- sys

SQL Editor

```
184
185
186 * SELECT *
187 FROM Account
188 WHERE Account_No = 10001;
189
190
191 * COMMIT;
192
193 * START TRANSACTION;
194
195 * UPDATE Account
196 SET Balance = Balance + 20000
197 WHERE Account_No = 10001;
198
199
```

Output

Action	Message	Duration
54 21:40:40 START TRANSACTION	0 row(s) affected	0.015 s
55 21:40:40 SELECT * FROM Account WHERE Account_No ...	1 row(s) returned	0.000 s
56 21:40:40 COMMIT	0 row(s) affected	0.000 s
57 21:41:04 START TRANSACTION	0 row(s) affected	0.000 s
58 21:41:04 SELECT * FROM Account WHERE Account_No ...	1 row(s) returned	0.000 s
59 21:41:05 COMMIT	0 row(s) affected	0.000 s
60 21:41:43 START TRANSACTION	0 row(s) affected	0.000 s
61 21:41:43 SELECT * FROM Account WHERE Account_No ...	1 row(s) returned	0.000 s
62 21:41:43 COMMIT	0 row(s) affected	0.000 s
63 21:42:09 START TRANSACTION	0 row(s) affected	0.000 s
64 21:42:09 UPDATE Account SET Balance = Balance + 200...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 s

Automatic context disabled. Use the manually get current caret position toggle automatic

49. WITHDRAWAL TRANSACTION

MySQL Workbench

2520090181_himasee

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

SCHEMAS

- 2520090181_himasee
- 2520090181_himasee_practical
- 2520090181_himasee_skill2
- Tables
- Views
- Stored Procedures
- Functions
- 2520090181_himasee_skill3
- 2520090181_himasee_skill3
- bank_acad_db
- Tables
- Views
- Stored Procedures
- Functions
- bank_db
- bookflow_db
- himasee_2520090181
- kludb
- sys

SQL Editor

```
192 * START TRANSACTION;
193
194 * UPDATE Account
195 SET Balance = Balance - 20000
196 WHERE Account_No = 10001;
197
198 * START TRANSACTION;
199
```

Output

Action	Message	Duration
56 21:40:40 COMMIT	0 row(s) affected	0.000 sec
57 21:41:04 START TRANSACTION	0 row(s) affected	0.000 sec
58 21:41:04 SELECT * FROM Account WHERE Account_No ...	1 row(s) returned	0.000 sec / 1
59 21:41:05 COMMIT	0 row(s) affected	0.000 sec
60 21:41:43 START TRANSACTION	0 row(s) affected	0.000 sec
61 21:41:43 SELECT * FROM Account WHERE Account_No ...	1 row(s) returned	0.000 sec / 1
62 21:41:43 COMMIT	0 row(s) affected	0.000 sec
63 21:42:09 START TRANSACTION	0 row(s) affected	0.000 sec
64 21:42:09 UPDATE Account SET Balance = Balance + 200...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
65 21:42:35 START TRANSACTION	0 row(s) affected	0.000 sec
66 21:42:35 SELECT Balance FROM Account WHERE Acco...	1 row(s) returned	0.000 sec / 1

Automatic context disabled. Use the manually get current caret position toggle automatic

50. CANCEL WITHDRAWAL

MySQL Workbench

2520090181_himasee

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

SCHEMAS

- 2520090181_himasee
- 2520090181_himasee_practical
- 2520090181_himasee_skill2
- Tables
- Views
- Stored Procedures
- Functions
- 2520090181_himasee_skill3
- 2520090181_himasee_skill3
- bank_acad_db
- Tables
- Views
- Stored Procedures
- Functions
- bank_db
- bookflow_db
- himasee_2520090181
- kludb
- sys

SQL Editor

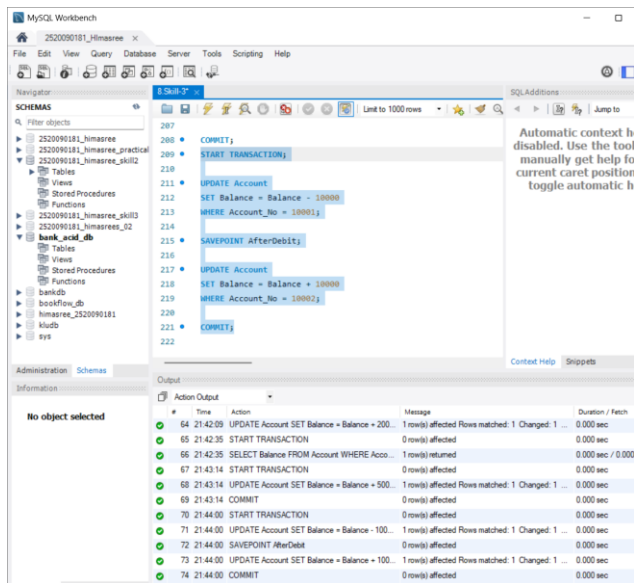
```
194 * SET Balance = Balance + 20000
195 WHERE Account_No = 10001;
196
197 * START TRANSACTION;
198
199 * SELECT Balance
200 FROM Account
201 WHERE Account_No = 10001;
202
203 * START TRANSACTION;
204
205 * UPDATE Account
206 SET Balance = Balance + 1000
207 WHERE Account_No = 10001;
208
209 * COMMIT;
210
```

Output

Action	Message	Duration
59 21:41:05 COMMIT	0 row(s) affected	0.000 sec
60 21:41:43 START TRANSACTION	0 row(s) affected	0.000 sec
61 21:41:43 SELECT * FROM Account WHERE Account_No ...	1 row(s) returned	0.000 sec / 0
62 21:41:43 COMMIT	0 row(s) affected	0.000 sec
63 21:42:09 START TRANSACTION	0 row(s) affected	0.000 sec
64 21:42:09 UPDATE Account SET Balance = Balance + 200...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
65 21:42:35 START TRANSACTION	0 row(s) affected	0.000 sec
66 21:42:35 SELECT Balance FROM Account WHERE Acco...	1 row(s) returned	0.000 sec / 0
67 21:43:14 START TRANSACTION	0 row(s) affected	0.000 sec
68 21:43:14 UPDATE Account SET Balance = Balance + 500...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
69 21:43:14 COMMIT	0 row(s) affected	0.000 sec

Automatic context disabled. Use the manually get current caret position toggle automatic

51. TRANSFER WITH SAVEPOINT



MySQL Workbench interface showing a SQL script for a bank transfer with a savepoint. The script is as follows:

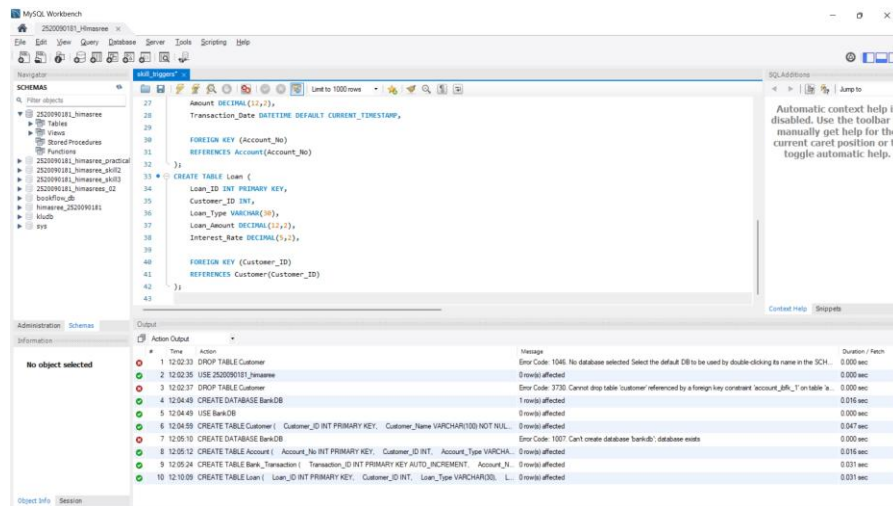
```
207 COMMIT;
208
209 START TRANSACTION;
210
211 UPDATE Account
212 SET Balance = Balance - 10000
213 WHERE Account_No = 10001;
214
215 SAVEPOINT AfterDebit;
216
217 UPDATE Account
218 SET Balance = Balance + 10000
219 WHERE Account_No = 10002;
220
221 COMMIT;
```

The output window shows the execution results:

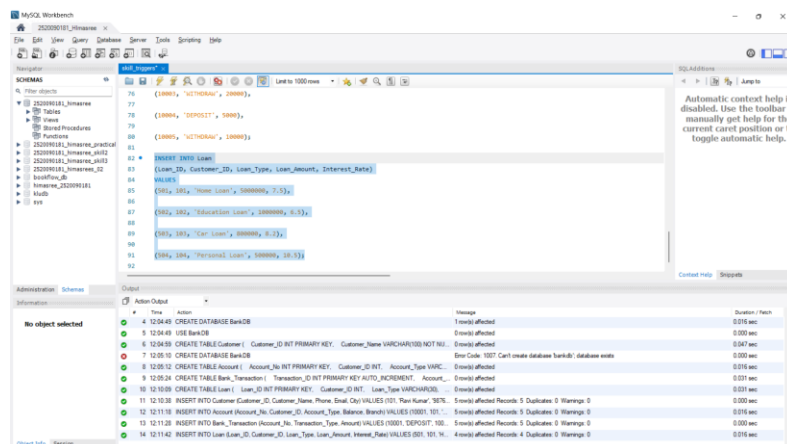
#	Time	Action	Message	Duration / Fetch
64	21:42:09	UPDATE Account SET Balance = Balance + 200...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
65	21:42:35	START TRANSACTION	0 row(s) affected	0.000 sec
66	21:42:35	SELECT Balance FROM Account WHERE Acco...	1 row(s) returned	0.000 sec / 0.000
67	21:43:14	START TRANSACTION	0 row(s) affected	0.000 sec
68	21:43:14	UPDATE Account SET Balance = Balance + 500...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
69	21:43:14	COMMIT	0 row(s) affected	0.000 sec
70	21:44:00	START TRANSACTION	0 row(s) affected	0.000 sec
71	21:44:00	UPDATE Account SET Balance = Balance - 100...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
72	21:44:00	SAVEPOINT AfterDebit	0 row(s) affected	0.000 sec
73	21:44:00	UPDATE Account SET Balance = Balance + 100...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
74	21:44:00	COMMIT	0 row(s) affected	0.000 sec

Skill-3_Triggers

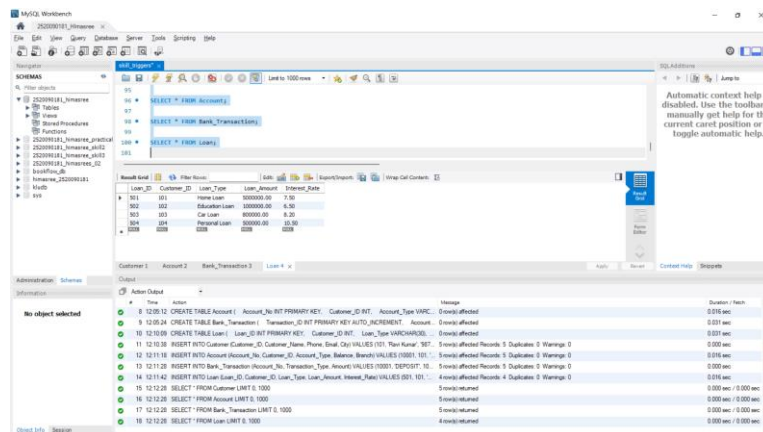
1.CREATE DATABASE AND TABLES



2.. INSERT DATA INTO TABLES



10. DISPLAY TABLE DATA



11. PROCEDURE TO DISPLAY ALL CUSTOMERS

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following code:

```
SELECT * FROM Customers;

DELIMITER ;

CALL GetAllCustomers();
```

The Results tab displays the output of the procedure, showing a table with 5 columns: Customer_ID, Customer_Name, Phone, Email, and City. The data is as follows:

Customer_ID	Customer_Name	Phone	Email	City
101	Ravi Kumar	9876543210	ravi@gmail.com	Hyderabad
102	Priya Sharma	9876543211	priya@gmail.com	Mumbai
103	Arun Reddy	9876543212	arun@gmail.com	Bangalore
104	Shruti Rao	9876543213	shruti@gmail.com	Chennai
105	Aman Kumar	9876543214	aman@gmail.com	Hyderabad

The Action Output tab shows the execution log, including the creation of the procedure and its successful execution.

12. PROCEDURE TO DISPLAY ACCOUNT DETAILS

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following code:

```
CALL GetAccountDetails(1001);
```

The Results tab displays the output of the procedure, showing a table with 5 columns: Account_No, Customer_ID, Account_Type, Balance, and Branch. The data is as follows:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad

The Action Output tab shows the execution log, including the creation of the procedure and its successful execution.

13. PROCEDURE TO FIND CUSTOMER ACCOUNTS

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following code:

```
WHERE C.Customer_ID = p_Customer_ID);

DELIMITER ;

CALL GetCustomerAccounts(101);
```

The Results tab displays the output of the procedure, showing a table with 5 columns: Customer_ID, Customer_Name, Account_No, Account_Type, Balance, and Branch. The data is as follows:

Customer_ID	Customer_Name	Account_No	Account_Type	Balance	Branch
101	Ravi Kumar	10001	Savings	50000.00	Hyderabad

The Action Output tab shows the execution log, including the creation of the procedure and its successful execution.

14. PROCEDURE TO DEPOSIT MONEY

The screenshot shows the MySQL Workbench interface with the 'SQL Editor' tab active. The SQL code defines a procedure named `DepositMoney` that takes two parameters: `p_Account_No INT` and `p_Amount DECIMAL(12,2)`. The procedure body includes a `BEGIN` statement, an `UPDATE Account` statement to increase the balance by `p_Amount` where `Account_No = p_Account_No`, an `END` statement, and a `DELIMITER ;` statement. The procedure is then called with `CALL DepositMoney(10001, 1000);`.

The 'Action Output' tab shows the execution results:

#	Time	Action	Message	Duration / Fetch
19	12:14:08	CREATE PROCEDURE GetCustomerDetails BEGIN SELECT * FROM Customer	Error Code: 1064: You have an error in your SQL syntax; check the manual that corresponds to your MySQL se...	0.000 sec
20	12:15:05	CREATE PROCEDURE GetCustomerDetails BEGIN SELECT * FROM Customer	1 row(s) affected	0.016 sec
21	12:15:05	CALL GetCustomerDetails(10001)	1 row(s) returned	0.000 sec / 0.000 sec
22	12:15:48	CREATE PROCEDURE GetAccountDetails IN p_Account_No INT BEGIN SELECT * FROM Account	1 row(s) affected	0.016 sec
23	12:16:44	CALL GetAccountDetails(10001)	1 row(s) returned	0.000 sec / 0.000 sec
24	12:16:29	CREATE PROCEDURE GetCustomerAccounts IN p_Customer_ID INT BEGIN SELECT * FROM CustomerAccounts	1 row(s) affected	0.000 sec
25	12:16:44	CALL GetCustomerAccounts(10001)	1 row(s) returned	0.000 sec / 0.000 sec
26	12:17:28	CREATE DATABASE BankDB	Error Code: 1007: Can't create database 'bankdb'; database exists	0.000 sec
27	12:17:35	CREATE PROCEDURE DepositMoney IN p_Account_No INT, IN p_Amount DECIMAL(12,2) BEGIN	1 row(s) affected	0.000 sec
28	12:17:35	CALL DepositMoney(10001, 1000)	1 row(s) affected	0.016 sec
29	12:17:40	CALL DepositMoney(10001, 1000)	1 row(s) affected	0.016 sec
30	12:18:32	CALL DepositMoney(10001, 1000)	1 row(s) affected	0.000 sec

15. PROCEDURE TO WITHDRAW MONEY

The screenshot shows the MySQL Workbench interface with the 'SQL Editor' tab active. The SQL code defines a procedure named `WithdrawMoney` that takes two parameters: `p_Account_No INT` and `p_Amount DECIMAL(12,2)`. The procedure body includes a `BEGIN` statement, an `UPDATE Account` statement to decrease the balance by `p_Amount` where `Account_No = p_Account_No`, an `END` statement, and a `DELIMITER ;` statement. The procedure is then called with `CALL WithdrawMoney(10001, 1000);`.

The 'Action Output' tab shows the execution results:

#	Time	Action	Message	Duration / Fetch
21	12:15:05	CALL GetCustomerDetails(10001)	1 row(s) returned	0.000 sec / 0.000 sec
22	12:15:48	CREATE PROCEDURE GetAccountDetails IN p_Account_No INT BEGIN SELECT * FROM Account	1 row(s) affected	0.016 sec
23	12:16:44	CALL GetAccountDetails(10001)	1 row(s) returned	0.000 sec / 0.000 sec
24	12:16:29	CREATE PROCEDURE GetCustomerAccounts IN p_Customer_ID INT BEGIN SELECT * FROM CustomerAccounts	1 row(s) affected	0.000 sec
25	12:16:44	CALL GetCustomerAccounts(10001)	1 row(s) returned	0.000 sec / 0.000 sec
26	12:17:28	CREATE DATABASE BankDB	Error Code: 1007: Can't create database 'bankdb'; database exists	0.000 sec
27	12:17:35	CREATE PROCEDURE DepositMoney IN p_Account_No INT, IN p_Amount DECIMAL(12,2) BEGIN	1 row(s) affected	0.000 sec
28	12:17:35	CALL DepositMoney(10001, 1000)	1 row(s) affected	0.016 sec
29	12:17:40	CALL DepositMoney(10001, 1000)	1 row(s) affected	0.016 sec
30	12:18:32	CALL DepositMoney(10001, 1000)	1 row(s) affected	0.000 sec
31	12:18:32	CALL WithdrawMoney(10001, 1000)	1 row(s) affected	0.000 sec

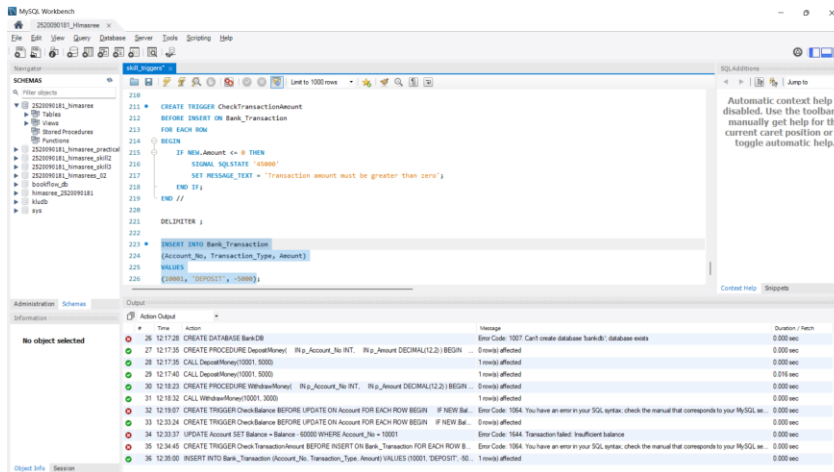
16. TRIGGER – PREVENT NEGATIVE BALANCE

The screenshot shows the MySQL Workbench interface with the 'SQL Editor' tab active. The SQL code defines a trigger named `CheckBalance` that fires BEFORE UPDATE ON Account. The trigger body includes a `BEGIN` statement, an `IF NEW.Balance < 0 THEN` condition, a `SIGNAL SQLSTATE '45000'` statement, a `SET MESSAGE_TEXT = 'Transaction failed: Insufficient balance';` statement, an `END IF;` statement, and an `END` statement. The trigger is then created with `CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN`.

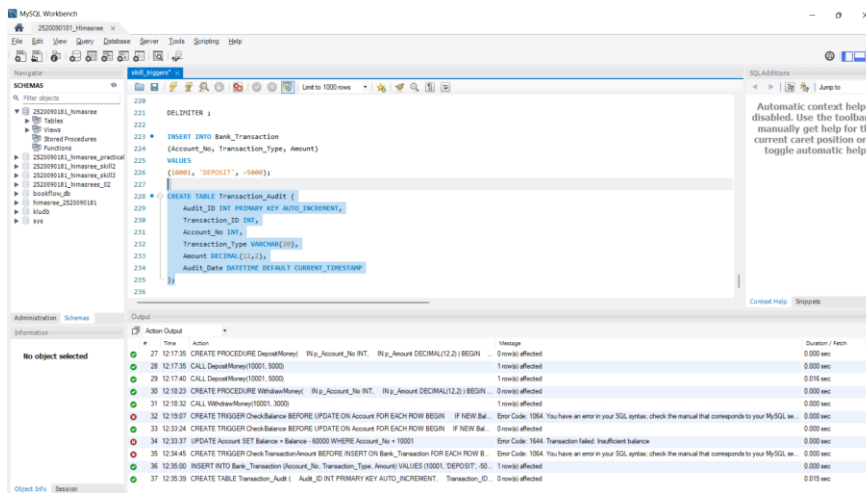
The 'Action Output' tab shows the execution results:

#	Time	Action	Message	Duration / Fetch
24	12:16:29	CREATE PROCEDURE GetCustomerAccounts IN p_Customer_ID INT BEGIN SELECT * FROM CustomerAccounts	1 row(s) affected	0.000 sec
25	12:16:44	CALL GetCustomerAccounts(10001)	1 row(s) returned	0.000 sec / 0.000 sec
26	12:17:28	CREATE DATABASE BankDB	Error Code: 1007: Can't create database 'bankdb'; database exists	0.000 sec
27	12:17:35	CREATE PROCEDURE DepositMoney IN p_Account_No INT, IN p_Amount DECIMAL(12,2) BEGIN	1 row(s) affected	0.000 sec
28	12:17:35	CALL DepositMoney(10001, 1000)	1 row(s) affected	0.000 sec
29	12:17:40	CALL DepositMoney(10001, 1000)	1 row(s) affected	0.016 sec
30	12:18:32	CREATE PROCEDURE WithdrawMoney IN p_Account_No INT, IN p_Amount DECIMAL(12,2) BEGIN	1 row(s) affected	0.000 sec
31	12:18:32	CALL WithdrawMoney(10001, 1000)	1 row(s) affected	0.000 sec
32	12:19:07	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN	IF NEW Bal... Error Code: 1064: You have an error in your SQL syntax; check the manual that corresponds to your MySQL se...	0.000 sec
33	12:23:24	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN	IF NEW Bal... Error Code: 1064: You have an error in your SQL syntax; check the manual that corresponds to your MySQL se...	0.000 sec
34	12:33:37	UPDATE Account SET Balance = Balance - 60000 WHERE Account_No = 10001	Error Code: 1644: Transaction failed; Insufficient balance	0.000 sec

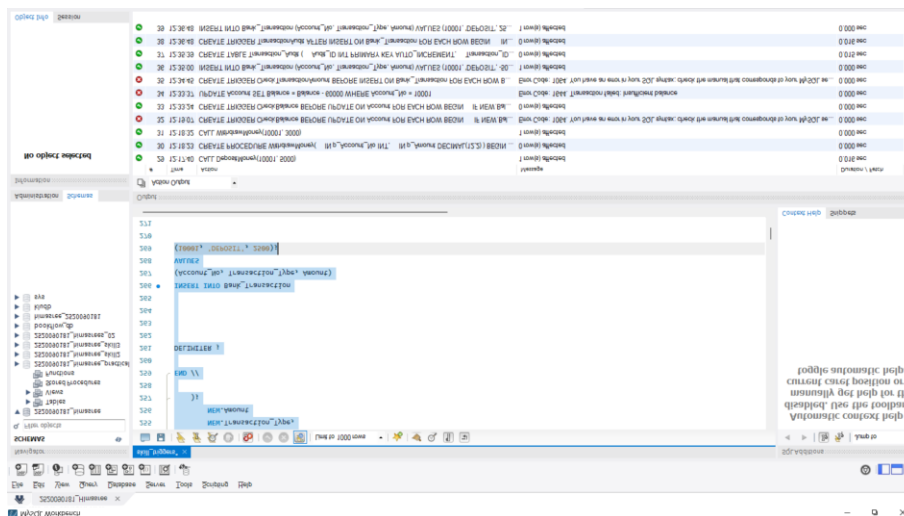
17. TRIGGER – PREVENT NEGATIVE DEPOSIT



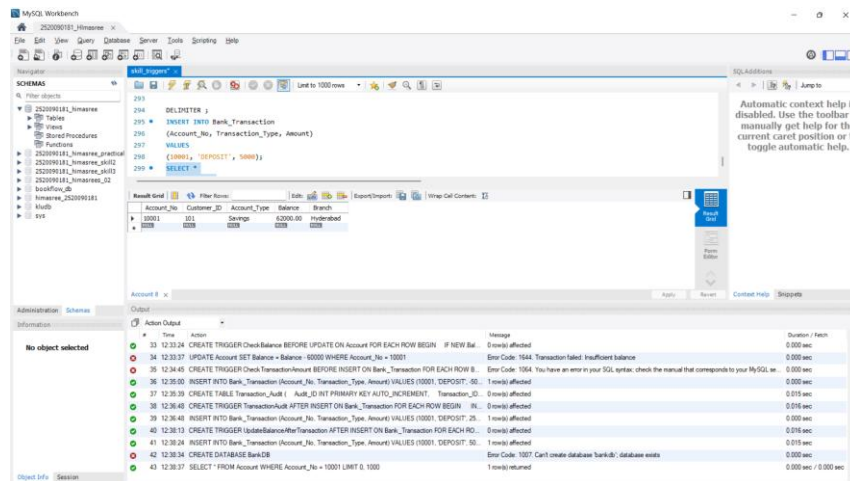
18. CREATE TRANSACTION AUDIT TABLE



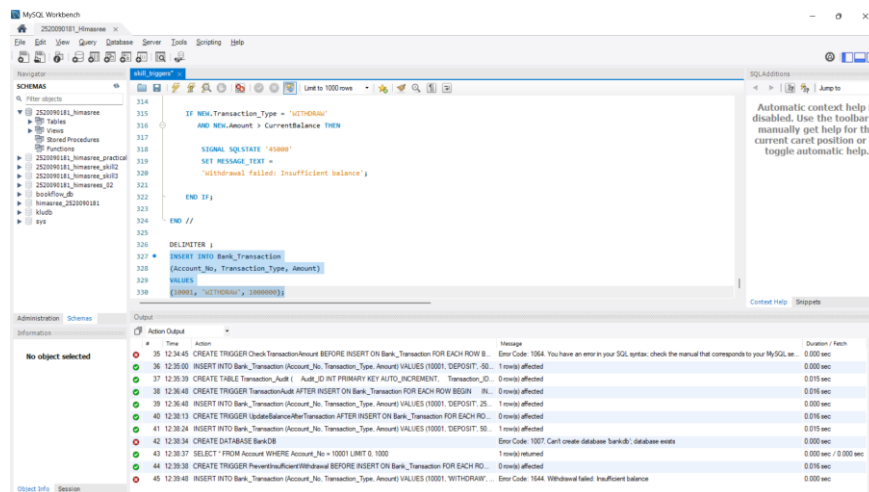
19. TRIGGER – TRANSACTION AUDIT



20. TRIGGER – AUTOMATICALLY UPDATE ACCOUNT BALANCE

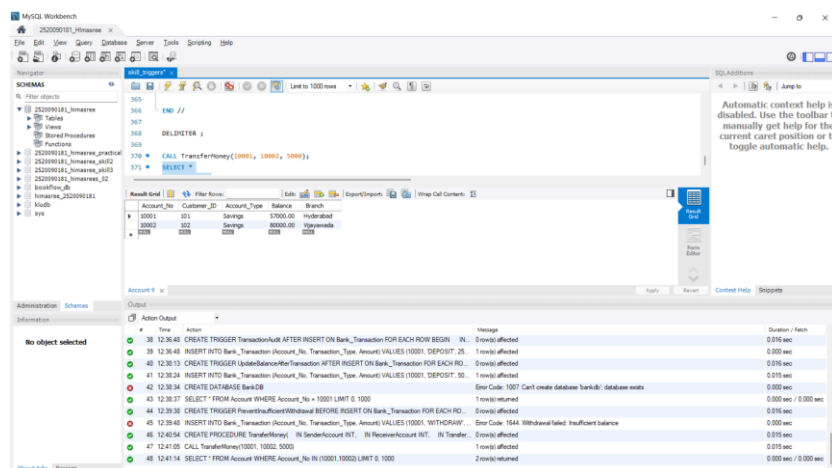


21. TRIGGER – PREVENT WITHDRAWAL ABOVE BALANCE



ADVANCED STORED PROCEDURES

22. PROCEDURE – ACCOUNT TRANSFER



23. PROCEDURE – CUSTOMER LOAN DETAILS

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following code:

```
DELIMITER ;
CALL GetCustomerLoans(101);
```

The Results tab displays the output of the procedure call:

Customer_Name	Loan_ID	Loan_Type	Loan_Amount	Interest_Rate
Ravi Kumar	101	Home Loan	900000.00	7.20

The Action Output tab shows the execution log:

#	Time	Action	Message	Duration / Fetch
1	13:28:48	CREATE DATABASE BankDB	Error Code: 1007: Can't create database 'bankdb': database exists	0:00 sec
2	13:28:51	USE BankDB	0 rows affected	0:00 sec
3	13:29:35	CREATE PROCEDURE GetCustomerLoans(IN p_Customer_ID INT) BEGIN SELECT C.Customer_N...	0 rows affected	0:00 sec
4	13:29:36	CALL GetCustomerLoans(101)	1 row(s) returned	0:00 sec / 0:00 sec

24. PROCEDURE – HIGH BALANCE ACCOUNTS

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following code:

```
WHERE Balance >= Min(Balance)
ORDER BY Balance DESC;
DELIMITER ;
CALL HighBalanceAccounts(50000);
```

The Results tab displays the output of the procedure call:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Current	120000.00	Bangalore
10005	105	Current	90000.00	Hyderabad
10002	102	Savings	80000.00	Vijayawada
10001	101	Savings	57000.00	Hyderabad

The Action Output tab shows the execution log:

#	Time	Action	Message	Duration / Fetch
1	13:28:48	CREATE DATABASE BankDB	Error Code: 1007: Can't create database 'bankdb': database exists	0:00 sec
2	13:28:51	USE BankDB	0 rows affected	0:00 sec
3	13:29:35	CREATE PROCEDURE GetCustomerLoans(IN p_Customer_ID INT) BEGIN SELECT C.Customer_N...	0 rows affected	0:00 sec
4	13:29:36	CALL GetCustomerLoans(101)	1 row(s) returned	0:00 sec / 0:00 sec
5	13:30:46	CREATE PROCEDURE HighBalanceAccounts(IN MinBalance DECIMAL(12,2)) BEGIN SELECT * ...	0 rows affected	0:016 sec
6	13:30:56	CALL HighBalanceAccounts(50000)	4 row(s) returned	0:000 sec / 0:000 sec

25. PROCEDURE WITH IN, OUT PARAMETERS

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following code:

```
FROM Account
WHERE Account_No = p_Account_No;
DELIMITER ;
CALL GetBalance(Limit, @CurrentBalance);
```

The Results tab displays the output of the procedure call:

Account_No	Customer_ID	Account_Type	Balance
10001	101	Current	120000.00

The Action Output tab shows the execution log:

#	Time	Action	Message	Duration / Fetch
1	13:28:48	CREATE DATABASE BankDB	Error Code: 1007: Can't create database 'bankdb': database exists	0:00 sec
2	13:28:51	USE BankDB	0 rows affected	0:00 sec
3	13:29:35	CREATE PROCEDURE GetCustomerLoans(IN p_Customer_ID INT) BEGIN SELECT C.Customer_N...	0 rows affected	0:00 sec
4	13:29:36	CALL GetCustomerLoans(101)	1 row(s) returned	0:000 sec / 0:000 sec
5	13:30:46	CREATE PROCEDURE HighBalanceAccounts(IN MinBalance DECIMAL(12,2)) BEGIN SELECT * ...	0 rows affected	0:016 sec
6	13:30:56	CALL HighBalanceAccounts(50000)	4 row(s) returned	0:000 sec / 0:000 sec
7	13:31:46	CREATE PROCEDURE GetBalance(IN p_Account_No INT, OUT p_Balance DECIMAL(12,2)) BEGIN	0 rows affected	0:015 sec
8	13:31:56	CALL GetBalance(1001, @CurrentBalance)	1 row(s) affected	0:000 sec
9	13:31:56	SELECT @CurrentBalance LIMIT 0, 1000	1 row(s) returned	0:000 sec / 0:000 sec